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Abstract

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Keywords: Economic sanctions, Human rights, International spillovers, Event study

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Tracing Sanctions through the Value Chain: Evidence from the US Cotton Ban ^{*}

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January 27, 2026

Abstract

We investigate to what extent and in what way the US Cotton Ban on products originating from Xinjiang province affected Chinese exports. We find that direct exports of cotton and cotton products to the United States fell by 19% at the intensive margin and 4% at the extensive margin. While the import reduction is strongest for Chinese provinces sourcing most actively from Xinjiang, reductions in US imports from third countries correlate only very weakly with the extent to which countries source cotton inputs from China. The European Union did not impose sanctions, but its cotton-related imports from China declined significantly as well. Our analysis demonstrates that this negative spillover effect is not due to lower prices, nor to EU countries indirectly facing the ban as they want to maintain access to the US market. The most likely reason for the import decline is European firms' concerns about reputational damage associated with using cotton products sourced from China.

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1 Introduction

Economic sanctions have become an important tool of foreign policy. They aim to change contested policies of other nations by imposing economic costs on them, while steering clear of military intervention.¹ The use of sanctions has tripled since the 1990s (Morgan, Syropoulos, and Yotov, 2023), including humanitarian sanctions aimed at curbing human rights violations and corruption. Sanctions have become increasingly targeted to minimize the repercussions for the population at large. Conventional economic sanctions are imposed on an entire nation, while targeted sanctions are imposed on specific individuals, businesses or organizations that are considered most responsible for the contested policies.

One example of such targeted, humanitarian sanctions is the import ban imposed by the United States in December 2020 on cotton originating from Xinjiang province. This is the largest cotton growing area in China, accounting for approximately 20 percent of global cotton production. In response to allegations of widespread repression of the Uyghur minority and the use of forced labor, the United States banned imports of all cotton and products containing cotton sourced from Xinjiang.² To further protest the use of forced labor in the region, it subsequently enacted the Uyghur Forced Labor Prevention Act (UFLPA) which broadened the ban to all products produced in Xinjiang, subject to a rebuttal procedure.

Despite the possibly wide-ranging implications of the US Cotton Ban, there is no evidence yet on its effectiveness. On the one hand, it is challenging for customs authorities to enforce such a region-specific policy in an industry that operates with a globally fragmented production structure (Gereffi, 1999). Its implementation requires detailed information on how the sanctioned products are embedded in local and global value chains. Hence, our first research question is whether it is possible to enforce the ban? On the other hand, the same interconnection makes it likely that downstream producers and countries operating in the same value chain are indirectly affected. It may make the sanctions less targeted, but it may also amplify their effects as it indirectly ensnares countries in between cotton-producers in Xinjiang and final consumers in the United States in the ban. The second research question then asks who is affected by the ban?

We focus on the impact of two key policy interventions that constitute the US Cotton Ban on Chinese exports: the withhold release order targeting Xinjiang-origin cotton implemented in December 2020 and the Uyghur Forced Labor Prevention Act (UFLPA) that was enacted

¹Notable instances include the multi-country sanctions against Iran’s nuclear program (Draca, Garred, Stickland, and Warrinnier, 2022) and Russia’s invasion of Ukraine (Ahn and Ludema, 2020; Crozet and Hinz, 2020; B  lin and Hanousek, 2021).

²The region-wide ban covers all products that contain cotton grown in Xinjiang—including textiles, garments and apparel—irrespective of the location of final production, within or outside of China.

in December 2021 and took effect in June 2022. The analysis uses product-level monthly trade flow between country pairs from UN Comtrade and similar trade flows between Chinese provinces and all country destinations from the General Administration of Customs of China. Affected cotton products are identified at the harmonized system (HS) 6-digit level of detail and control products are non-cotton textile and apparel products in the same HS 2-digit categories.

The baseline model estimates the average effect of the US sanctions on Chinese exports using a triple-differences strategy with a rich set of interactive fixed effects. It compares the evolution of Chinese exports to the United States relative to a product-destination specific baseline, controlling for product-month and destination-month interaction fixed effects that flexibly capture supply and demand shocks. The baseline estimates indicate that the Cotton Ban had a statistically and economically significant negative effect on Chinese exports of cotton-containing products to the United States. Specifically, we find a 19% decline in export value at the intensive margin and a 4% reduction in the likelihood that a product is exported (3 percentage points extensive margin effect).

These results beg the question whether this decline is large or small. It far exceeds the direct flow of goods from Xinjiang to the United States prior to the ban, but the reduction falls far short of a complete ban on imports from other Chinese provinces. Using trade information and the Chinese Input-Output table both at the provincial level, we investigate whether enforcement is aligned with input linkages between Chinese provinces and Xinjiang. We find that the export reductions line up well with the extent to which the textile sectors in various Chinese provinces source agricultural and textile inputs from Xinjiang.³ Direct exports from Xinjiang are effectively curtailed (-82%) and the probability of direct exports falls to near zero. By far the largest other decline is in provinces in the top quartile, with a smaller, but still significant decline for provinces in the third quartile. However, this pattern does not replicate when we perform the same analysis on US imports from other countries. Depending on the specification used, there is only a weak relationship or none at all between a country's decline in US exports and the intensity of its cotton and cotton products sourcing from China. US customs seems to be able to trace the cotton's origin through the Chinese value chain, but unable or unwilling to trace Chinese-sourced cotton in value chains that span third countries.

We also find indirect effects of the US Cotton Ban on exports to other destinations. Given that cotton production in Xinjiang remained stable, producers needed to find other markets.⁴

³We use predetermined input-output linkages to classify all provinces in four quartiles.

⁴Statistics from China's National Bureau of Statistics and the International Cotton Advisory Committee indicate that the Xinjiang share of global cotton production remained stable after 2020 at approximately

However, the value of exports from China to the EU also declined (-9%), while effects at the extensive margin are negative for several countries with the magnitudes broadly in line with countries’ political alignment with the United States. We investigate several possible explanations for the EU’s import decline in spite of not imposing sanctions on China itself. This negative spillover effect is not due to greater buying power that could have reduced prices, nor to EU countries with a US-oriented textile sector self-imposing a ban to maintain export access to the US market. We find that the import decline is most pronounced in EU countries with the greatest human rights awareness, suggesting that the most likely explanation is European firms’ concerns about reputational damage if they were to sell cotton products sourced from China. The fact that the EU import decline is largest for finished apparel products, where importers deal directly with final consumers, supports this interpretation. This pattern contrasts with the US import decline being most pronounced for raw cotton products, where the Xinjiang origin is easiest to establish.

Related literature The first literature we contribute to studies targeted sanctions. Draca et al. (2022) evaluate the effectiveness of sanctions imposed on Iran’s political elite during the negotiations over its nuclear program. They find a positive response to news of negotiation breakthroughs on the returns of targeted entities on the Tehran Stock Exchange, indicative of economic incentives for the local elites to negotiate a deal in exchange for sanctions relief. A few studies look at the impact of US and EU sanctions on Russian firms following the annexation of Crimea in 2014. Ahn and Ludema (2020) find that they harmed the financial performance of targeted Russian firms, but also that strategic companies were less affected. In contrast, Nigmatulina (2023) finds no systematic negative financial effects on targeted companies, but provides evidence of substantial shielding of state-owned firms by the Russian government. Maystadt, Tsang, Van Biesebroeck, and Warrinnier (2024) study financial sanctions imposed by the US government on the Xinjiang Production and Construction Cooperation. In contrast with these studies which study sanctions targeting specific firms, we study sanctions that target specific products and trace the effects through the value chain.

A second related literature studies the relationship between sanctions and trade performance. Not surprisingly, bilateral trade declines when one country imposes sanctions on another. This has been shown at the national level (e.g., Gutmann, Neuenkirch, and Neumeier, 2023), firm level (e.g., Haidar, 2017), and product level (e.g., Bělin and Hanousek, 2021; Mironova, 2023). More broadly, we contribute to the body of work exploring the connection between political conflicts and trade. Conflict with China has been very prominent and neg-

21% (Figure 1).

ative effects on trade have been associated with the US-China trade war (Fajgelbaum and Khandelwal, 2022), consumer boycotts (Heilmann, 2016), political conflict (Li, Jian, Tian, and Zhao, 2021), or diplomatic rows over visits by the Dalai Lama (Fuchs and Klann, 2013). Our contribution is to take global value chain (GVC) connections explicitly into account. Makioka and Zhang (2024) is one of the few studies that considers the impact of trade conflict on sourcing, which is quite natural given their focus on export restrictions. We show that GVC considerations make it more challenging to enforce sanctions, but also lead to indirect effects on third countries.

The indirect effects that we document for the EU in particular contribute to an emerging literature on political consumerism that spans economics, marketing, and political science. The existing evidence indicates that consumers are more likely to purchase from sellers who share a common political affiliation or a common position on controversial social issues (McConnell, Margalit, Malhotra, and Levendusky, 2018; Schoenmueller, Netzer, and Stahl, 2023; Liaukonytė, Tuchman, and Zhu, 2023; Conway and Boxell, 2024). One manifestation of such preferences are consumer boycotts, but the evidence on their trade effects remains mixed. Koenig and Poncet (2022) document that French textile firms reduced sourcing from Bangladesh after the Rana Plaza collapse due to concerns over reputational damage. Our results suggest that EU firms consider the reputational risk of sourcing cotton products from Xinjiang to be sufficiently high to reduce their imports from China even in the absence of any formal EU sanctions.

The remainder of the paper is organized as follows. Section 2 provides background information on the implementation of the US Cotton Ban and Section 3 describes the data. Next, we discuss the direct effects, i.e., the impact of the ban on the export of Chinese cotton products to the United States, in Section 4 and the indirect effects on other countries in Section 5. Section 6 concludes.

2 Background

2.1 Cotton production in Xinjiang

Benefiting from favorable climate and policy support from the Chinese government, Xinjiang has become the main cotton production base in China (Gale and Davis, 2022).⁵ Its output expanded significantly between 2010 and 2019, from 2.5 to 5 million tons per year. Its share

⁵Government support includes production subsidies for cotton cultivation and transport subsidies for textile manufacturers in other regions that source cotton from Xinjiang (US Department of Agriculture, 2025).

of total Chinese production also doubled, growing from 42% to 85%, and Xinjiang alone accounted for one-fifth of global cotton production which totaled 25 million tons in 2019 (Voora, Bermudez, Farrell, Larrea, and Luna, 2023). Total cotton production in Xinjiang was valued at 78.8 billion RMB, or approximately 6% of Xinjiang GDP.⁶ The Xinjiang Production and Construction Corps (XPCC), a paramilitary organization known for its extensive use of mechanized farming techniques, is the region’s largest cotton producer (Carlson and Weaver, 2022).⁷

Upland cotton, which is suitable for textile and garment production, is the dominant variety cultivated in Xinjiang. Long-staple cotton, which is regarded as higher quality due to its longer fibers and used for fine fabrics, is expanding but represents only a small fraction of output. Most of the raw cotton produced in Xinjiang is processed into textiles and garments in manufacturing hubs such as Jiangsu, Zhejiang, and Shandong, while only a small share is exported in raw form (Carlson and Weaver, 2022). Through the export of Chinese cotton-related products, Xinjiang cotton is closely integrated into global supply chains.

In 2019, China’s exports of cotton-related products reached a total value of 40.1 billion USD, which represented 25% of global exports. Fabrics and finished consumer goods account for 97.6% of this, and it is estimated that approximately half of all exports contain cotton originating from Xinjiang (Carlson and Weaver, 2022).⁸ These products are generally not exported directly from Xinjiang, but leave China from other provinces and are sometimes re-exported via third countries before arriving at their final destinations (Lehr and Bechrakis, 2019). Complex and opaque supply chain links make it difficult for custom officials, regulators and stakeholders to trace the origin of cotton products.

2.2 Forced labor in Xinjiang

Since 2017, the Xinjiang government has faced accusations of repressing Uyghurs and other ethnic minorities in the region. The allegations include the establishment of detention camps and the extensive use of surveillance against minorities. In particular, the production of cotton and cotton products in Xinjiang has long been subject to international suspicion of using forced labor from ethnic minority groups, particularly the Uyghurs. These allegations have been documented extensively in media reports,⁹ by non-governmental organizations

⁶This value is obtained by multiplying the quantity of production by the average cotton price in China.

⁷In 2019, the XPCC accounted for 41% of Xinjiang’s cotton production and 34% of the total for China.

⁸Trade statistics are based on the UN Comtrade database and the Xinjiang share is estimated for 2021.

⁹The *New York Times* in particular has actively reported on this issue, going back to December 16, 2018 “China’s Detention Camps for Muslims Turn to Forced Labor”, with major investigative journalism pieces

(e.g., Lehr, 2020; C4ADS, 2021), and by academic researchers (e.g., Murphy et al., 2022; Zenz, 2023). The US government (Carlson and Weaver, 2022) and the United Nations (OHCHR, 2022) have published the results of their investigations in official reports. The Chinese government has protested the legality of the UN investigation and it has consistently denied all allegations, claiming specifically that no instances of coerced labor exist in cotton production or for any other goods produced in Xinjiang province.¹⁰

Despite objections from the Chinese government, concerns over forced labor practices in Xinjiang have continued to gain traction in Western countries. Human rights organizations have advocated boycotts of products linked to Xinjiang cotton. They have criticized the sourcing practices of global apparel brands and retailers,¹¹ and have called for formal sanctions against Xinjiang cotton.¹²

2.3 The US Cotton Ban

Despite repeated denials by the Chinese government, the United States imposed financial, investment and trade sanctions against Chinese entities and senior Chinese officials for human rights abuses in Xinjiang. Maystadt et al. (2024) evaluate the effects of various firm-level sanctions, but here we focus only on the trade effects of the region-wide Cotton Ban.

In September 2019, the US Customs and Border Protection (CBP) issued nine withhold release orders (WROs) to prevent importation of goods produced by Chinese entities which it considered associated with forced labor in Xinjiang. These initial actions focused on specific companies rather than imposing a broad ban.¹³ Additionally, on July 1, 2020, four US government departments—State, Treasury, Commerce, and Homeland Security—jointly issued the Xinjiang Supply Chain Business Advisory, warning US firms of forced labor in

published, for example, on April 8, 2022 “Supply Chains Widely Tainted by Forced Labor in China, Panel Is Told” and as recent as May 29, 2025 “How China Uses Work to Reshape Uyghur Identity and Control a Strategic Region”.

¹⁰For instance, Yilijiang Anayiti, the spokesperson of the Xinjiang government, asserted during a press conference in Beijing on April 9, 2021 that there were no instances of forced labor in Xinjiang, and that the employment of ethnic minorities was entirely voluntary. Press release available at http://de.china-embassy.gov.cn/zt/zgxxj/202104/t20210415_9045932.htm. The Chinese government also released an official rebuttal to the OHCHR (2022) report available online at https://www.ohchr.org/sites/default/files/documents/countries/2022-08-31/ANNEX_A.pdf.

¹¹For example, in July 2020, more than 180 organizations urged apparel brands to cease sourcing cotton from Xinjiang. Press release available at <https://enduyghurforcedlabour.org/news/402-2/>.

¹²For example, Human Rights Watch (2021) calls for new legislation to prohibit all imports linked to forced labor.

¹³The CBP already considered implementing a region-wide import ban in September 2020, but this proposal was opposed by the Trump administration. Instead, the CBP issued six additional WROs targeting specific entities. Reuters report is available at <https://www.reuters.com/world/asia-pacific/us-pulls-back-broad-import-bans-cotton-tomatoes-chinas-xinjiang-2020-09-15/>.

the Xinjiang cotton sector and urging them to conduct enhanced due diligence.

After further negotiations with stakeholders, the United States implemented broader import sanctions targeting all cotton and cotton products originating in Xinjiang. Figure 2 outlines the key legislative actions in the implementation of the US Cotton Ban. The first broad-based measure was announced on December 2, 2020 when the CBP issued a WRO prohibiting the importation of all cotton and tomato products produced by the XPCC or any of the affiliated entities of its sprawling network.¹⁴ This order authorizes the CBP to detain imports of intermediates and final goods such as textiles, garments, and apparel at US ports of entry if there is reasonable evidence that the goods were produced, in whole or in part, with XPCC-origin cotton. Detained shipments can only be released if the importer provides proof that the products were not made under such conditions. Subsequently, on January 13, 2021, the CBP issued a second, even more expansive region-wide WRO, extending the ban to all cotton and tomato products produced in Xinjiang, regardless of the producer.¹⁵

To further strengthen efforts against the use of forced labor in the region, one year later the United States escalated its trade restrictions by enacting the Uyghur Forced Labor Prevention Act (UFLPA). The Act was signed into law by President Biden on December 23, 2021 and took effect, with immediate enforcement, on June 21, 2022. The UFLPA differs from the WRO regime in two key respects.

First, its enforcement framework is more stringent, requiring that importers provide “clear and convincing evidence” to CBP within a 30-day period to substantiate that shipments are free from forced labor. The evidentiary requirement is significantly more demanding than the documentation standards under the WRO. Required documentation includes a detailed supply chain map delineating all involved entities, information on workforce practices at each stage (including recruitment processes, wage payments, and per-worker production output), and “credible audits” evaluating forced labor risks and remediation measures (US Customs and Border Protection, 2022).

Furthermore, the Act mandates the Forced Labor Enforcement Task Force (FLETF) to develop and implement a strategy to strengthen enforcement (US Department of Homeland Security, 2022). This strategy includes: (i) conducting a detailed assessment of the scope of imports associated with forced labor; (ii) establishing an official UFLPA Entity List and identifying high-priority sectors for enforcement; (iii) strengthening the capacity to trace and

¹⁴The effective date of this order was backdated to November 30, 2020. Press release available at <https://www.dhs.gov/news/2020/12/02/dhs-issues-detention-order-cotton-products-made-xinjiang-production-and-construction>.

¹⁵Press release available at <https://www.cbp.gov/newsroom/national-media-release/cbp-issues-region-wide-withhold-release-order-products-made-slave>.

identify goods linked to forced labor; (iv) enhancing the legal authorities and enforcement tools at US ports of entry; (v) issuing guidance for importers to ensure compliance; and (vi) fostering interagency coordination and consultation.

Second, the coverage of the UFLPA is substantially broader. It establishes a rebuttable presumption that all goods produced wholly or partially in Xinjiang, or by entities on the UFLPA Entity List, are made with forced labor. As a result, all such shipments—extending beyond cotton and tomato products—need in principle be detained at US borders and denied entry.

Of course, enforcing such a broad ban is extremely difficult in a world where production of many goods has fragmented geographically and final goods imported from anywhere can contain inputs sourced from Xinjiang. The FLETF has designated the cotton, tomato, and poly-silicon industries as high priority sectors with dedicated enforcement plans. We therefore examine the impact of trade sanctions on products containing Xinjiang-origin cotton. We define a fixed set of cotton-related products as the targeted commodities for the entire sample period (as well as a fixed set of control products), despite the broader scope of the UFLPA. In the post-UFLPA period, we are implicitly evaluating the effect of the FLETF priority strategy. To assess the robustness of our findings, we implement alternative definitions of both targeted and control product groups in supplementary analyses.

3 Data

3.1 National trade data from UN Comtrade

For the country-level analysis, we rely on trade flows from the United Nations Comtrade database that are defined at the bilateral country level by product and month. The sample period always runs from January 2019 to December 2022. Where possible, we measure trade flows from the import side which tend to be recorded more accurately than exports.

The sample includes all products defined at the 6-digit level of the Harmonized System (HS) commodity classification that are textiles or textile products. We define them as the 801 products contained in the HS 2-digit categories from 50 to 65 (excluding 64). Following Carlson and Weaver (2022), we identify 238 cotton-related products that are sanctioned; the remaining 563 products are the controls. They include raw cotton products (5 sanctioned out of a total of 29), yarn products (49 of 150), fabrics (126 of 336), apparel products (47 of 211), and other textile products (11 of 78).¹⁶

¹⁶We use the correspondence tables of the UN Statistics Division to make the product classification consis-

In the benchmark analysis, the sample includes Chinese exports to 126 destination countries with positive imports of at least one of the sample commodities in 2019. EU member states are included as separate countries. Hong Kong and Macau are excluded as they are Special Administrative Regions of China. When we investigate the heterogeneity of effects across countries that differ in their sourcing of cotton inputs from China, we use two different samples. First, we combine exports from China with twenty additional origin countries into a four-dimensional panel. Second, we use information on US imports from 186 countries with positive textile and apparel trade, for the same period and for a subset of 798 products showing positive US imports. Appendix D contains additional details.

The final sample for the benchmark analysis is a balanced panel with 4,844,448 observations in total, comprising 801 products exported to 126 destinations over a 48-month period.¹⁷ Summary statistics for the targeted and control commodities during a 12-month period in 2019 (prior to the Cotton Ban) are presented in Table 1, separately for the five product groups. Three quarters of the sanctioned product-month observations (2,141 of 2,856) show positive US imports and their average (monthly) value was 3.5 million USD. Positive imports in the EU and other control destinations—which tend to be much smaller markets than the United States—are less likely (37%) and on average of much lower value.

3.2 Provincial trade data from China Customs

Not all cotton-related exports from China were subject to sanctions, only those using Xinjiang-origin cotton. To investigate the impact of the US Cotton Ban specifically on exports from Xinjiang and from Chinese provinces that source most intensively from Xinjiang, we use information on trade flows at the provincial level. These data are collected by twenty-nine regional customs offices and reported to China’s General Administration of Customs. Consistent with the UN Comtrade data, we focus on export flows between January 2019 and December 2022, to the same 126 destination countries, and of the same 796 products (for five products we observe no records in the China Customs dataset).

The resulting four-dimensional balanced panel comprises 139,612,032 product-province-destination-month observations. 5,011,355 observations, approximately 3% of the sample, show positive exports. Table 2 contains summary statistics in 2019 for all provinces (Panel A), for Xinjiang (Panel B), and for the top quartile of provinces according to their sourcing

tent over time. Table A.1 in the Appendix shows the breakdown of treated and control products by 2-digit category. Three control products are included in both the yarn and fabrics groups.

¹⁷Twenty-eight destination countries stopped reporting trade flow data in the last years of the sample period. Rather than assigning no trade in these instances, we rely on export data reported by China. Treating these instances as missing data produces similar results (available upon request).

from Xinjiang (Panel C). The average (monthly) export value across all Chinese provinces of to-be-sanctioned products to the United States was 136,000 USD, to EU members states 4,000 USD, and for other control observations (products and/or destinations) 6,000 USD. Direct exports from Xinjiang to the United States or the EU were strikingly low, while exports from Chinese provinces that rely most on Xinjiang cotton inputs were much higher than average. A similar pattern is observed for the probability of exporting to-be-sanctioned products. The average export probability to the United States for all Chinese provinces was 16%, but only 1% for Xinjiang and 31% for top-quartile provinces.

3.3 Other datasets

Interprovincial Input-Output table in China. We use the Chinese interprovincial Input-Output table to identify Chinese provinces that most intensively source cotton from Xinjiang. It is compiled by Chen et al. (2023) using a trade matrix derived from Chinese VAT invoices to calibrate the multi-regional Input-Output table provided by the Development Research Center of the State Council. The resulting table provides an overview of the interprovincial supply chain linkages among 42 industrial sectors in China in 2017. In Appendix B, we detail how we rank all provinces according to their intensity of cotton-related input sourcing in their textile-related sectors. This provides an approximation to the extent of exposure to the US Cotton Ban for each province.

Google Trends data. We use keyword search intensity data from Google Trends to rank EU countries according to their level of interest in or awareness of human rights issues. This is accomplished by measuring the relative online search intensity for a set of “topics” (keywords) that are related to historical human rights crises from the 2010s. In Appendix C, we provide further details about the methodology and the country ranking generated by the aggregate index.

4 Direct effects on US imports

4.1 Baseline specification

We estimate the direct impact of the US Cotton Ban on the exports of Chinese cotton products using the triple-differencing strategy of Frazer and Van Biesebroeck (2010):

$$Y_{pdt} = \beta Ban_{pdt}^{US} + \gamma_{pd} + \gamma_{pt} + \gamma_{dt} + \epsilon_{pdt}. \quad (1)$$

The units of observation are product-destination-months (pdt) and the specification includes three sets of interaction fixed effects. It focuses identification on changes over time within the cross-sectional product-destination pairs, while controlling for product-month fixed effects to capture supply shocks and destination-month fixed effects as controls for demand shocks.¹⁸ The indicator Ban^{US} takes on the value of one for US imports of cotton-related products from December 2020 onward. The coefficient β will turn negative if Chinese exports of cotton products to the United States decline after the US Cotton Ban is enacted relative to the trend for non-cotton products exported from China to the United States, further normalizing the cotton versus non-cotton trend for US exports by the same relative trend in export growth to other, non-sanctioned countries. Standard errors are clustered at the product-destination level, consistent with the level of treatment assignment.

The dependent variable Y_{pdt} is the logarithm of export volume when we study the effects at the intensive margin, and it is a binary indicator for positive export when we study the extensive margin effects. We also show robustness checks using the Poisson pseudo maximum likelihood (PPML) estimator that accounts for zeros and heteroskedasticity in the residuals.

To identify a causal effect, US sanctions should be allocated exogenously relative to the products' potential export evolution. Unlike sanctions of Chinese high-tech products that are imposed for economic reasons, the US Cotton Ban is mainly motivated by humanitarian and political considerations and can be considered exogenous (Ahn and Ludema, 2020; Nigmatulina, 2023). The comprehensive set of fixed effects capture unobserved heterogeneity that may be correlated with the allocation of sanctions.

A valid diff-in-diffs estimator requires comparable pre-sanction trends between the target and control product-destination pairs for both outcome variables (Olden and Møen, 2022). We test the parallel trends assumption by estimating the following event study:

$$Y_{pdt} = \sum_{t=2019m1, t \neq 2020m11}^{2022m12} \beta_t Ban_{pdt}^{US} + \gamma_{pd} + \gamma_{pt} + \gamma_{dt} + \epsilon_{pdt}, \quad (2)$$

where November 2020, the final month of the pre-treatment period, is the base period with β_t set to zero.

¹⁸The product-month fixed effects also remove potential spillovers from the Cotton Ban on control commodities.

4.2 Baseline results

Table 3 shows the estimates of the average impact of the region-wide Cotton Ban. The estimate in column (1) indicates that exports of sanctioned commodities from China to the US decline by 19% (point estimate of -0.211 in log points) relative to the trends of the control group. This effect is significant at the 1% level. Considering that the total annual export volume of all Chinese cotton-containing products to the United States in 2019 was 10 billion USD, the aggregate reduction of Chinese exports was 1.9 billion USD per year.

The estimate in column (3) shows the extensive margin effect, i.e., the extent to which cotton-related HS 6-digit products have a reduced probability of positive exports from China to the United States after the Cotton Ban is enacted. The point estimate is again negative and statistically significant at the 1% level. The probability of any export of Chinese cotton-containing products decreased by 3 percentage points, which is a 4% reduction from the average export probability of 75% in 2019.

Dynamic impact. As discussed previously, the Cotton Ban gradually tightened restrictions on Chinese cotton imports. Using the same four-year sample period, we divide the 25-month treatment window into three distinct phases: (1) December 2020–November 2021 represents the first year when the ban was implemented by a WRO of the CBP; (2) December 2021–June 2022 covers most of the second year when the UFLPA was passed, but not in force yet; and (3) July 2022–December 2022 is the period following the implementation of the UFLPA. We interact the Ban^{US} indicator with separate interaction effects for these three periods.

Columns (2) and (4) in Table 3 show the corresponding point estimates. The magnitude of the sanction effects progressively increase over time (in absolute value). At the intensive margin, the export decline grows from -16% (-0.169) in the initial year to -28% (-0.331) under the UFLPA. The gradual increase in the negative effect on the probability of Chinese exports is even more pronounced, with the point estimate tripling from -0.02 to -0.07 over the sanction period. This pattern may be attributed to adjustments of supply chains by American importers taking some time, but also to improved enforcement and greater restrictiveness of the Cotton Ban under the UFLPA.

Parallel trends. We test the parallel trends assumption for both outcome variables. The two panels of Figure 3 plots all estimated coefficients for 47 month-specific interaction terms with the 95% confidence intervals. There are no statistically significant export differences between Chinese exports of cotton products to the United States and the control product-

destination pairs prior to the treatment period. Tests of joint-significance of all pre-sanction coefficients can be rejected both for the export volume and the export dummy. The reduction in export volumes and export probability during the sanction period shows up in the red markers. The three periods considered in Table 3 are indicated by dashed vertical lines, and the gradual increase of the export decline, in particular at the extensive margin, is clearly visible. In the final period, all but one point estimate are significantly different from zero.

Robustness checks. We verify the robustness of the results using a Poisson Pseudo Maximum Likelihood estimator (PPML), which is robust to the presence of zero-value observations and to heteroskedastic errors (Silva and Tenreyro, 2006). The dependent variable is the export volume in levels and results are reported in Table E.1 in the Appendix. In column (1), the average point estimate for the US Cotton Ban effect remains negative and statistically significant at the 1% level and with a value of -0.105 it lies in between the effects estimated at the intensive and extensive margins. As in the baseline, the absolute magnitude of the effect progressively increased over time.¹⁹

Furthermore, we verify whether results are robust to using information on trade flows from China’s Customs data. Results in columns (1) and (2) of Table E.2 rely on the full database that has four dimensions, as total Chinese exports for each product-destination-month observation are reported separately for each Chinese province (indexed by o for origin). Note that we do not include product-destination-month fixed effects in the regression as this would be fully collinear with the Cotton Ban indicator, Ban^{US} . The point estimates measure the average effect for all Chinese provinces and are negative and statistically significant at both the intensive and extensive margin. The absolute magnitude of the export decline is lower than in the benchmark case, but we turn to heterogeneity across Chinese provinces in the next section.

Results in columns (3) and (4) of Table E.2 aggregate the Chinese Customs data to the national level which generates a dataset with the same dimensions as the UN Comtrade data used for the baseline results. The main distinction is that export transactions are now measured at the export rather than the import side. The intensive margin export decline is estimated quite a bit larger at -26%, while the extensive margin results become a lot smaller and statistically insignificant.

¹⁹We implement the estimator with the Stata routine “ppmlhdfc” of Correia et al. (2020). It automatically drops observations that do not contribute to the identification, e.g., destination-product pairs that never show positive exports, which explains the lower number of observations compared to the extensive margin regressions.

4.3 Enforcement of the US Cotton Ban through the value chain

Chinese supply chain. We already established that the Cotton Ban reduced US imports of cotton-related products from China. However, the ban did not target all Chinese products, but only those made with cotton sourced from Xinjiang. Such targeted sanctions intend to minimize the impact on other products. The bilateral country data from UN Comtrade cannot fully capture the effectiveness of Cotton Ban enforcement, which is why we now turn to China’s Customs data which contain information on Chinese exports at the province level.

We expect the export decline to be strongest for direct exports from Xinjiang and to be increasing in the extent to which other provinces’ textile industry sources cotton from Xinjiang. Using the interprovincial Input–Output table of Chen et al. (2023), we classify Chinese provinces into four quartiles based on their dependence on cotton-related inputs from Xinjiang.²⁰ Figure B.1 in the Appendix shows the relevant input values for all Chinese provinces. Major textile and garment production hubs such as Guangdong, Zhejiang, and Hebei, depend heavily on cotton-related inputs from Xinjiang and are included in the top quartile $Q4$.

We then examine the effectiveness of the Cotton Ban using the following specification:

$$Y_{podt} = Ban_{pdt}^{US} \times \left[\beta^{XJ} XJ_o + \sum_{k=1}^4 \beta^{Qk} \{Qk \text{ of XJ-sourcing}\}_o \right] + \gamma_{pod} + \gamma_{pot} + \gamma_{dt}/\gamma_{pdt} + \epsilon_{podt}. \quad (3)$$

Observations are indexed by product p , provincial origin o , export destination d , and month t . XJ is an indicator for Xinjiang province and the terms in brackets are four quartile indicators for intensity of sourcing from Xinjiang. Specification (3) allows the identification of five coefficients. We can make the specification even more flexible by replacing the destination-month fixed effect with product-destination-month triple-interaction fixed effects. In that case, we can only estimate four of the five coefficients, as this is now a fully saturated model at the province (origin) level, and the point estimates measure the difference with the excluded quartile $Q1$.²¹

²⁰The procedure is described in detail in Appendix B. Tibet is excluded as it is also subject to US allegations of human rights abuses; Beijing is excluded as it may confound the estimation given that it is the national capital and political center, but only a small textile production center.

²¹While econometrically feasible, we do not include province–destination–month fixed effects. In that case, the effects would be only identified from the extent to which differences in the export evolution for cotton and non-cotton products within province-destination pairs are related to each province’s extent of sourcing from Xinjiang. The problem with this identification strategy is that many provinces are relatively specialized in their exports to some destinations. When they export almost exclusively either cotton or non-cotton textile products, there is no identifying variation left.

Table 4 presents the estimation results for the two variations of equation (3). At the intensive margin, the direct effect of the sanctions is always estimated by far the largest specifically for Xinjiang cotton exports, and much larger than the overall effect for China from the baseline. The point estimate of -1.105 in column (1) implies that cotton exports decline by two thirds under the Cotton Ban. The point estimate of -1.732 in column (3) implies that the export level of Xinjiang is reduced by 83% relative to exports of provinces in the first quartile.²²

Most importantly, we also find large and statistically significant declines in exports to the United States from Chinese provinces that source most cotton from Xinjiang, i.e., those in the fourth and third quartiles. The magnitudes of all effects increase monotonically with the degree of dependence on Xinjiang’s inputs, indicating that the cotton ban led to larger export reductions in provinces more reliant on Xinjiang-sourced inputs.

Estimates at the extensive margin, in columns (2) and (4), are also largest for Xinjiang, but the differences with the coefficients for other provinces are notably smaller. However, the point estimate of -0.024 is extremely large compared to the mean probability of cotton-related products exported from Xinjiang to the United States which was only 1% in 2019. It suggests that the Cotton Ban stopped virtually all direct exports from Xinjiang to the United States. The export probability declined most for other provinces that are in the third quartile—not the fourth—but differences in the baseline probabilities may explain the uneven effects.

Global supply chain. We next perform a similar analysis investigating the impact on exports from a broad sample of countries. The above results indicate that US Customs are able to deter exports systematically more from Chinese provinces that rely more strongly on Xinjiang inputs. Given that Xinjiang produces one fifth of global cotton output and that the Chinese textile industry is the largest in the world, many other countries also rely directly or indirectly on cotton that originates from Xinjiang. Are US imports also systematically reduced for them?

We investigate this question symmetrically to specification (3). Rather than using 29 Chinese provinces as origins, we now use export information for a sample of 21 (including China) countries from the UN Comtrade dataset. All selected countries are among the 55 most important exporters of textile products to the United States, but they are chosen to

²²Export transactions reported by Xinjiang still indicate 154 instances of cotton-related product shipments to the United States even after the implementation of the Cotton Ban, with a total value of 13 million USD. These transactions may reflect misreporting by Chinese Customs, e.g., misclassifications of product or destination, it may be goods that are detained at the US border, or transactions that are not intercepted.

make sure they vary in the extent to which they source cotton or cotton-products from China. We detail in Appendix D the sample of exporters and the construction of four sourcing quartiles using the reliance on Chinese cotton inputs of each origin country by its total volume of cotton products imported from China. Countries in the top quartile are most dependent on Chinese cotton and some of their US exports will no longer be permissible under the Cotton Ban. This group includes major textile and garment producers such as Vietnam, India, and Indonesia. The indicator Ban_{pdt}^{US} now switches to one for US imports of cotton products irrespective of the exporting country, such that the $(Ban_{pdt}^{US} \times \text{China})$ interaction equals the original Ban^{US} indicator and plays the same role as the $(Ban^{US} \times XJ)$ interaction in equation (3).

Table 5 presents the corresponding results. Two observations stand out. First, at both margins, the Cotton Ban reduces the import of Chinese cotton products. These direct sanction effects are estimated larger but comparable to the baseline estimates in Table 3. Second, the sanctions appear to have little impact on imports from other origin countries that rely the most on Chinese cotton inputs. The estimated effects for countries in $Q3$ and $Q4$ are statistically insignificant at the intensive margin, whereas at the extensive margin only $Q3$ and $Q1$ show a negative, statistically significant, but small effect. The estimates in column (3) which show the export evolution for the upper 3 quartiles relative to the first quartile show the expected negative signs, but point estimates are still small, especially compared to the corresponding estimates in column (3) of Table 4, and not statistically significant. Overall, it suggests that the Cotton Ban does not effectively restrict imports of cotton products that are made with Chinese cotton inputs but shipped from third countries.

Results in Table E.3 in the Appendix show somewhat stronger effects when the sample is expanded to 27 origin countries, which include all top-30 cotton exporters to the United States that report the necessary data. While US imports of countries in $Q4$ and $Q3$ do not fall, see column (1), they do not increase like the US imports of countries in $Q1$. As a result, the normalized effects in column (3) are negative. Extensive margin effects are negative for three of the four quartiles, but the effect is stronger in $Q3$ and $Q1$ than in $Q4$, thus not in line with the intensity of sourcing from China. Finally, in Table E.4 we show results based only on US imports from all 186 origin countries in a triple-difference specification. Here the extensive margin results are in line with the sourcing intensity, but the intensive margin effects not at all.

Overall, US import reductions under the Cotton Ban seem to be much less systematically related to supply chain exposure for imports from third countries, compared to imports from Chinese provinces.

5 Indirect effects on third country imports

5.1 Specification and results

Specification. Because many countries operate in the global value chain that produces textile products for the US market, the Cotton Ban may influence Chinese exports to other countries and regions as well. On the one hand, (mostly) Chinese products with Xinjiang-sourced cotton can no longer be exported to the United States and their producers will look for alternative destinations. On the other hand, textile output of other countries can no longer contain Xinjiang-sourced inputs if they are to be exported to the United States. Producers that want to retain that option may need to look for alternative cotton input sources.

The net effect on third countries can be estimated straightforwardly by adding a $\beta^C Ban_{pdt}^D$ term in equation (1), in addition to the original Ban_{pdt}^{US} indicator. The new indicator takes a value of one for cotton-related exports from China to country D during the period the Cotton Ban was in effect. The coefficient β^D captures any potential (net) indirect effects of the sanctions on country C .

Indirect effects on third countries. We report the estimates of indirect effects on Chinese exports to a select set of destination countries in Figure 4. It shows the point estimates with the corresponding 95% confidence interval for the intensive margin effect in panel (a) and the extensive margin effect in panel (b). For comparison purpose, we show the US estimates from Table 3 on the right side (in red). Countries are ordered by effect size, averaged over the two margins.

The volume of Chinese cotton exports to Russia grows significantly and China also starts exporting additional products to Russia. At the other extreme, cotton-related exports decline significantly to the EU on both margins, even though the EU did not impose any formal sanctions itself on Chinese products. At the intensive margin, no other point estimate is significantly different from zero, but the probability that China exports a cotton-containing product also declines for the UK, Canada, Brazil, and Japan.

The ordering of countries in Figure 4 seems to reflect not only economic integration, but is relatively similar to countries' geopolitical alignment, i.e, whether countries are more likely to side with China or the United States. For example, with the exception of Russia, the ordering is similar to the extent that countries' voting at the United Nations General Assembly aligns with China or the United States. Between 2017 and 2018, that showed the following similarity order: China, South Africa, India, Brazil, Mexico, Russia, Japan, EU,

Canada, UK, United States.

Indirect effects on the EU. We zoom in on the indirect effects of the US Cotton Ban on European imports of Chinese cotton-containing products. The estimates in columns (1) and (3) of Table 6 replicate the EU effects shown in Figure 4. The export reductions for the EU are, respectively, 60% and 56% smaller than for the United States, but highly significant. Even more than for the United States, the effects grow over time as the UFLPA is voted and then takes effect. The third coefficient is on both margins two to three times larger (in absolute value) than the immediate impact.

The event study for the indirect effects on EU imports, in Figure 5, indicates that the parallel trends assumption for the pre-treatment period is satisfied. The gradually increasing effect in the post-sanctions period also shows up very clearly, especially at the extensive margin.

Robustness checks We conduct a range of sensitivity checks and the results prove highly robust, i.e., consistently showing a significant decline of EU imports from China that is smaller than the direct effects on US imports (full results available upon request).

First, we estimate with a broader control group that includes non-textile and non-garment products. Second, we use a narrower control group that omits non-textile products that are ever exported directly from Xinjiang to the United States, because all Xinjiang products are sanctioned in the last seven months. Third, we rely on a more limited set of 98 destination countries as controls, because only they report any textile imports from China in every month throughout the sample period.

Next, we use alternative definitions for the treatment group. First, we aggregate all EU imports into a single entity which generated a remarkably similar, but less precise, point estimate of -0.101*** at the intensive margin. However, the estimated coefficient at the extensive margin becomes near zero. Second, using a more narrow definition of 195 instead of 238 treated commodities, following Carlson and Weaver (2022), barely changes the results.

Finally, the PPML estimate for the coefficient on Ban^{EU} is -0.072**, which is statistically significant only at the 5% level, and approximately half as large as the -0.129 direct effect on Ban^{US} .

5.2 Potential explanations

The indirect effects of the US Cotton Ban cannot be attributed to EU policy interventions, as no formal sanctions targeting Chinese commodities were implemented by the EU during the sample period. Supply-side factors, such as technological shocks affecting Chinese production, are controlled for through product-month fixed effects and, if anything, would lead us to expect increased EU imports through trade diversion. We explore three demand-side mechanisms that could potentially explain the import decline.

Reduced market power (of China). The US Cotton Ban lowered demand for Chinese cotton-containing products in the United States and in countries exporting final products to the United States. Given the size of the US market, this would reduce any market power Chinese exporters might have in other destinations and could put downward pressure on prices. If quantities adjust only slowly, lower prices could reduce the value of exports to other destination markets, such as the EU.

This can be straightforwardly tested by estimating equation (1) separately using log price and log quantity as dependent variables. Results in Table 7 clearly indicate that this mechanism cannot explain the EU import decline. The indirect effect is entirely driven by a reduction in quantity with no significant effect of sanctions on Chinese export prices.

Export potential to the United States. The US Cotton Ban requires all firms that export to the United States, not only Chinese firms, to provide sufficient evidence that none of their own cotton inputs or those of their suppliers are sourced from Xinjiang. As we discussed, the burden of proof increased substantially under the UFLPA. Faced with uncertainty regarding the traceability of cotton origins within Chinese supply chains, European firms, particularly those relying most on the US export market, may opt to reduce imports of cotton-containing products from China.²³ Both the need to avoid Xinjiang-origin cotton and the mere cost of compliance with the US Cotton Ban, could motivate reduced imports from China.

To empirically assess whether the re-exporting channel is a plausible explanation for the negative indirect effect, we divide all EU countries in two groups according to the extent their exports were directed to the United States prior to the Cotton Ban. Figure D.2 in Appendix D shows the full ranking and the six EU countries in the top quartile of US exposure, i.e., Portugal, Italy, Romania, and Malta, Lithuania, Latvia, and Bulgaria are categorized as

²³These incentives extend to firms that wish to maintain uniform product content across all consumer markets they serve.

“High” exporters to the US.

We incorporate an additional interaction term ($Ban^{EU} \times \text{High US exports}$) in equation (1) to capture any incremental effect of the Cotton Ban on these countries. Columns (1) and (3) of Table 8 show an unexpected positive effect at the intensive margin, and no statistically significant effect at the extensive margin. The EU import decline of Chinese cotton-containing products is clearly not driven by countries most reliant on the US export market.

This finding is not altogether unexpected. It is in line with the results in Table 5 where we also did not find lower US imports from countries that sourced cotton inputs most intensely from China. A robustness check verifying whether a pattern exists between US imports and specifically EU countries’ reliance on cotton imports from China, does not support this mechanism either.

Reputational concerns. While allegations of human rights violations in Xinjiang have been circulating since 2017, the US Cotton Ban could make the connection between Xinjiang cotton and the use of forced labor more salient for European consumers. Some consumers may become more conscientious about the origins of the products they purchase even without a formal ban. International non-governmental organizations have been instrumental in lobbying for the Cotton Ban in the United States. In Europe they did not succeed in getting formal sanctions approved, but they still appeal directly to consumers. Therefore, continuing to sell products made with cotton from China in markets with a heightened awareness of human rights issues could pose a risk to a company’s reputation and brand valuation.

We examine this mechanism by leveraging variation across EU countries in awareness about human rights issues. We utilize Google Trends data to construct such an awareness index for all EU countries, based on search activity surrounding historical human rights crises.²⁴ Details of the calculations are provided in Appendix C and Figure C.1 shows the results, with Germany, the Netherlands, Finland and Sweden showing the highest values. Among the nineteen EU countries that we can rank, those with a value above the median are classified as “High HR awareness.”²⁵

Estimation results with an additional interaction term for EU countries with high human rights awareness are also in Table 8. At the intensive margin, the coefficient of interest is estimated very precisely and equal to -0.153*** in column (2). With the interaction

²⁴We selected human right crises not involving China to avoid a possible endogeneity problem as anti-Chinese sentiments might affect trade with China, independently from the US Cotton Ban.

²⁵Due to data limitations, we cannot calculate the index for seven EU countries. Rather than omit them from the regression, we set the indicator for “High HR awareness” to zero for them.

term included, the baseline effect on Ban^{EU} shrinks almost to zero and loses statistical significance. It implies that the reduction in EU imports is driven entirely by countries with greater attention to human rights issues. This finding is consistent with Koenig and Poncet (2022) who also find that firms reduce sourcing from origin countries that pose reputational risks. However, this pattern does not replicate in the extensive margin results which show a small positive effect on the interaction term. This may be explained with reputational concerns chiefly affecting the behavior of large firms which dominate the impact at the intensive margin, but not at the extensive margin.

In Figure 6 we present one final piece of evidence that is consistent with the importance of reputational concerns. We classify all sample commodities into five subgroups and estimate the direct and indirect effects for the United States and the EU separately by product group. The point estimates for the direct impact on US imports (in red) are larger (in absolute value) for less processed cotton products. This is especially apparent at the extensive margin. It is intuitive that effects on imports that fall directly under the Cotton Ban legislation are strongest for products where the legal ban is easiest to enforce, such as raw, unprocessed cotton.

The indirect impact on EU imports (in blue) shows the reverse pattern. Effects at the intensive margin, which are dominated by large firms, are strongest for apparel and only slightly smaller for fabric. In contrast, there is no significant impact on yarn or raw cotton, and those point estimates are even positive. This pattern is consistent with indirect effects stemming from the reputation mechanism. In the EU, products closer to consumers see a larger decline; in the US, products that are less processed, and where Xinjiang origin is likely to be easier to detect, see a larger decline.

6 Conclusion

There has been a notable increase in the use of targeted sanctions, but the impact of such measures remains underexplored, particularly in the context of global value chains. We examine the region-wide ban on the imports of Xinjiang-origin cotton, implemented by the United States in December 2020, as a case study to provide new empirical evidence on the effectiveness of this policy tool.

Two findings stand out. First, it is challenging to implement such a targeted policy in a world with internationally fragmented production. When processed goods arrive at US ports, US customs needs to ascertain whether they contain Xinjiang-origin cotton or not. We find that the heterogeneity in the US import decline across Chinese provinces is correlated with

provinces' reliance on Xinjiang-origin cotton, which we verify using China's interprovincial Input-Output table. However, we fail to detect a systematic correlation for the change in US imports across countries, even though countries' reliance on Chinese-origin cotton also varies greatly.

Second, while the policy consists of targeted sanctions, sourcing links in global value chains potentially lead to spillover effects on other countries. While this can enhance the impact of the sanctions, as producers in other countries indirectly help enforce the US Cotton Ban, it may also spread the costs to producers in other countries. We investigate specifically for the EU which factors best explain the observed decline of cotton-related imports from China. Consumer-facing firms turning away from China in order to avoid potential damage to their reputation seems to be even more important than global value chain connections.

Our analysis also suggests a few areas for future research. While our identification strategy isolates the direct and spillover effects of the ban on Chinese exports, it does not speak to broader general-equilibrium effects. In particular, we do not observe how Chinese authorities and firms adjust along other margins, e.g., they might reduce Chinese own cotton imports, redirect Xinjiang-linked cotton products to domestic markets, shift sourcing between provinces, or use fiscal and industrial policies to cushion affected producers. These questions are important for understanding the full welfare and policy implications of the sanctions. While we document indirect effects of the Cotton Ban on Chinese exports to other countries, we do not explicitly analyze trade diversion and how sanctions lead to a reorganization of the global textile value chain, e.g., which countries gain? We also do not explore the potential consequences of the Cotton Ban for the United States itself, the country imposing the sanctions. Specifically, the adjustment in global value chains could yield both positive and negative economic outcomes for American firms and consumers. To investigate these issues, a more structural model-based approach is likely to be required.

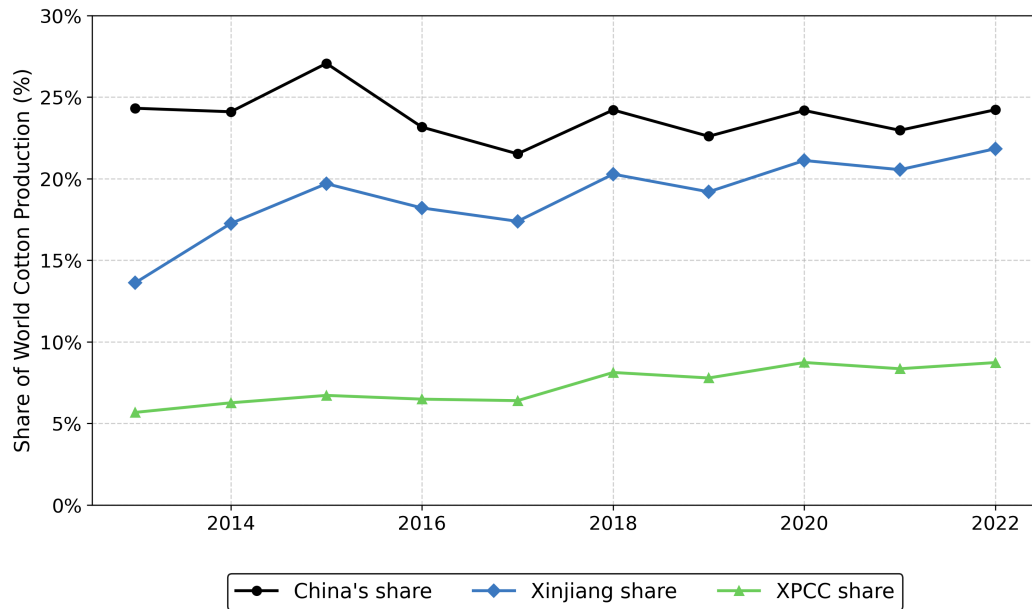
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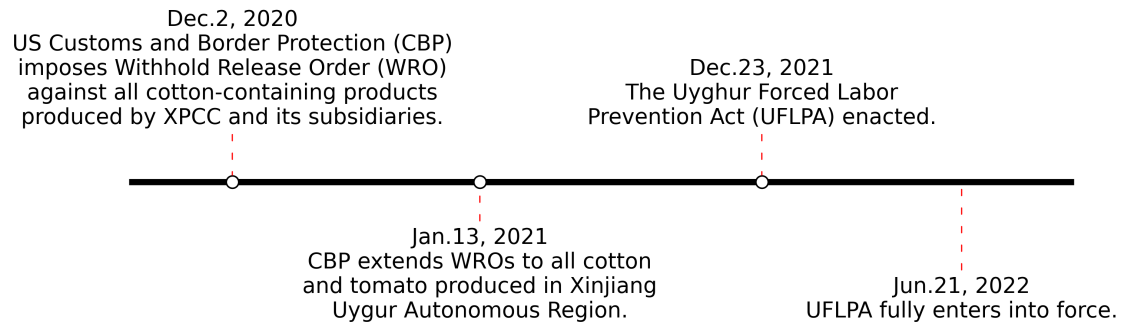
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Figure 1: China's share in world cotton production



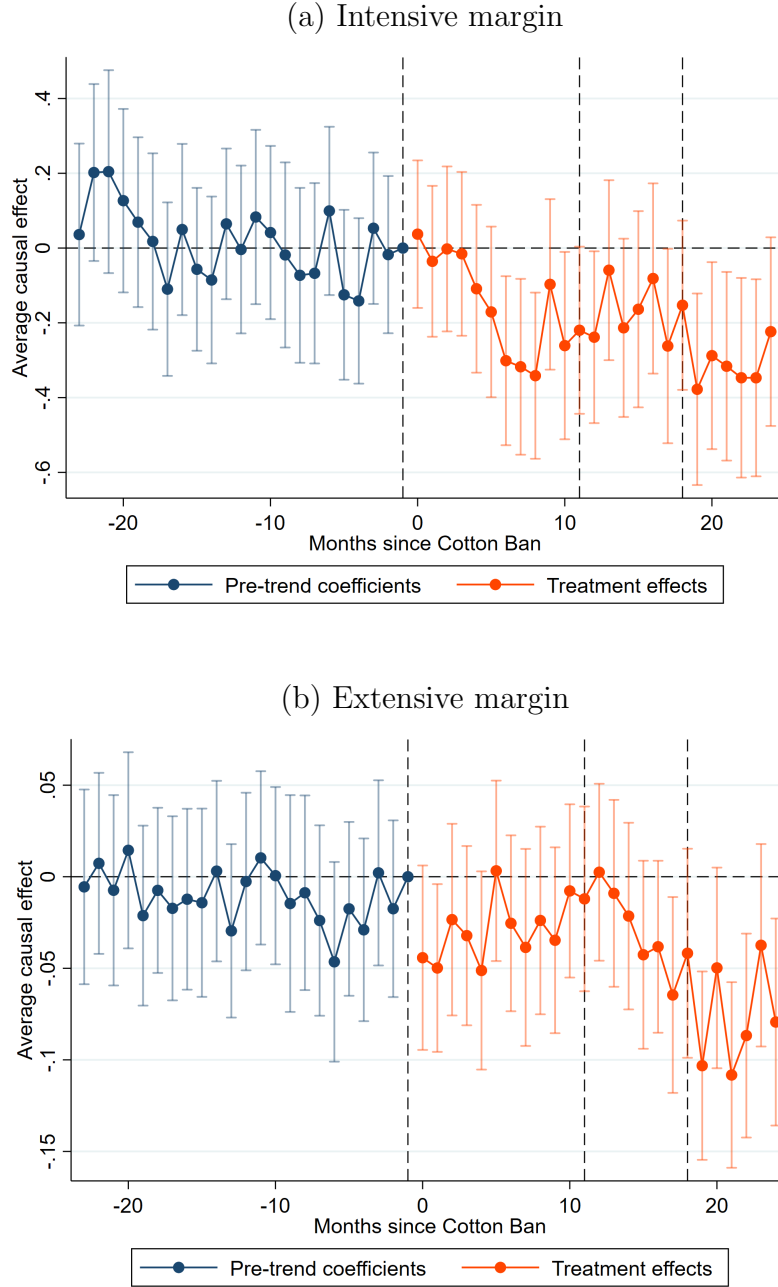
Source: Compiled by authors based on information from the National Bureau of Statistics of China and International Cotton Advisory Committee.

Figure 2: Timeline for the imposition of the US Cotton Ban



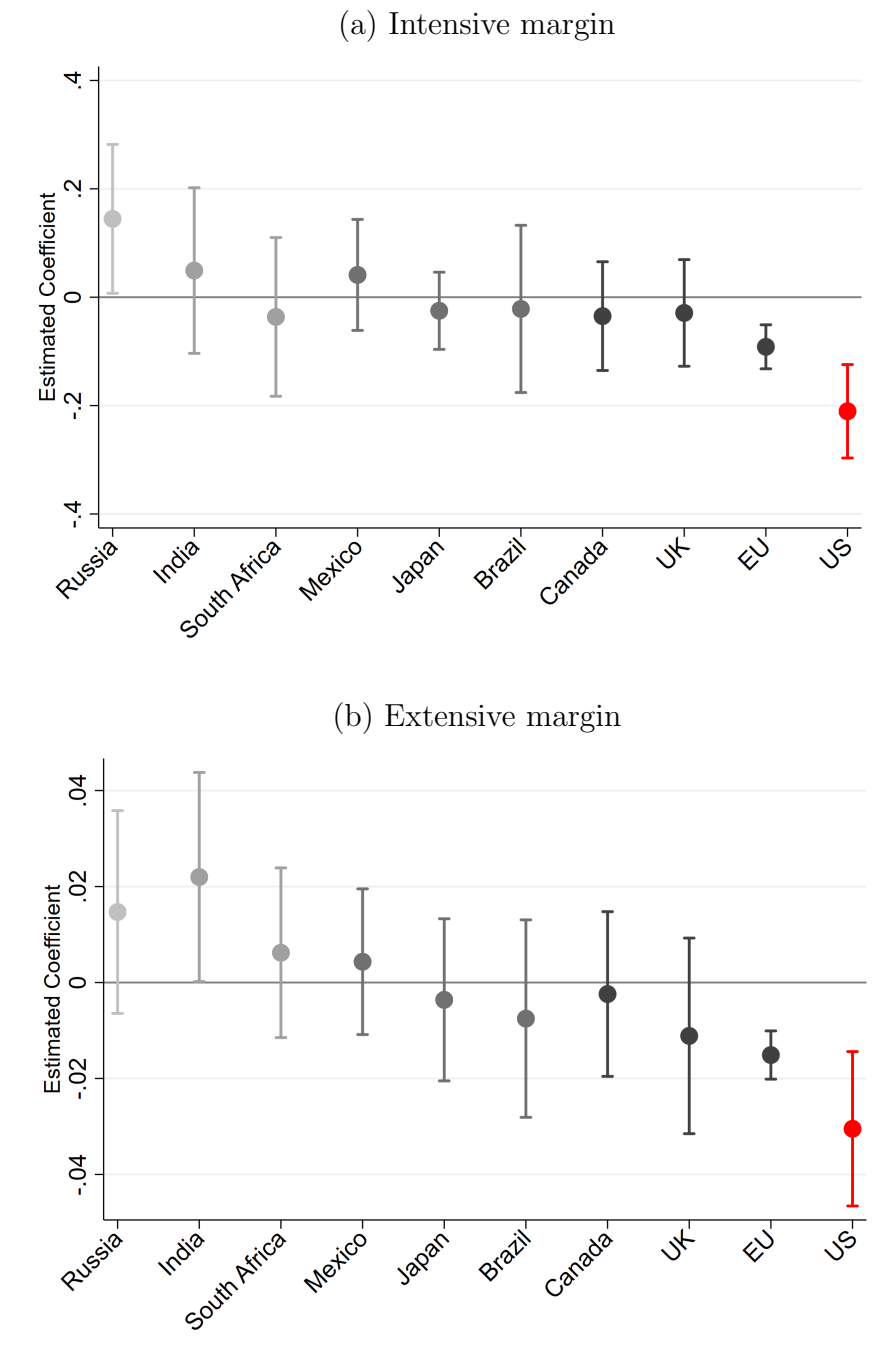
Source: Information from the US Customs and Border Protection (2023), accessible at <https://www.cbp.gov/trade/forced-labor/withhold-release-orders-and-findings>.

Figure 3: Event study of the direct impact on US imports



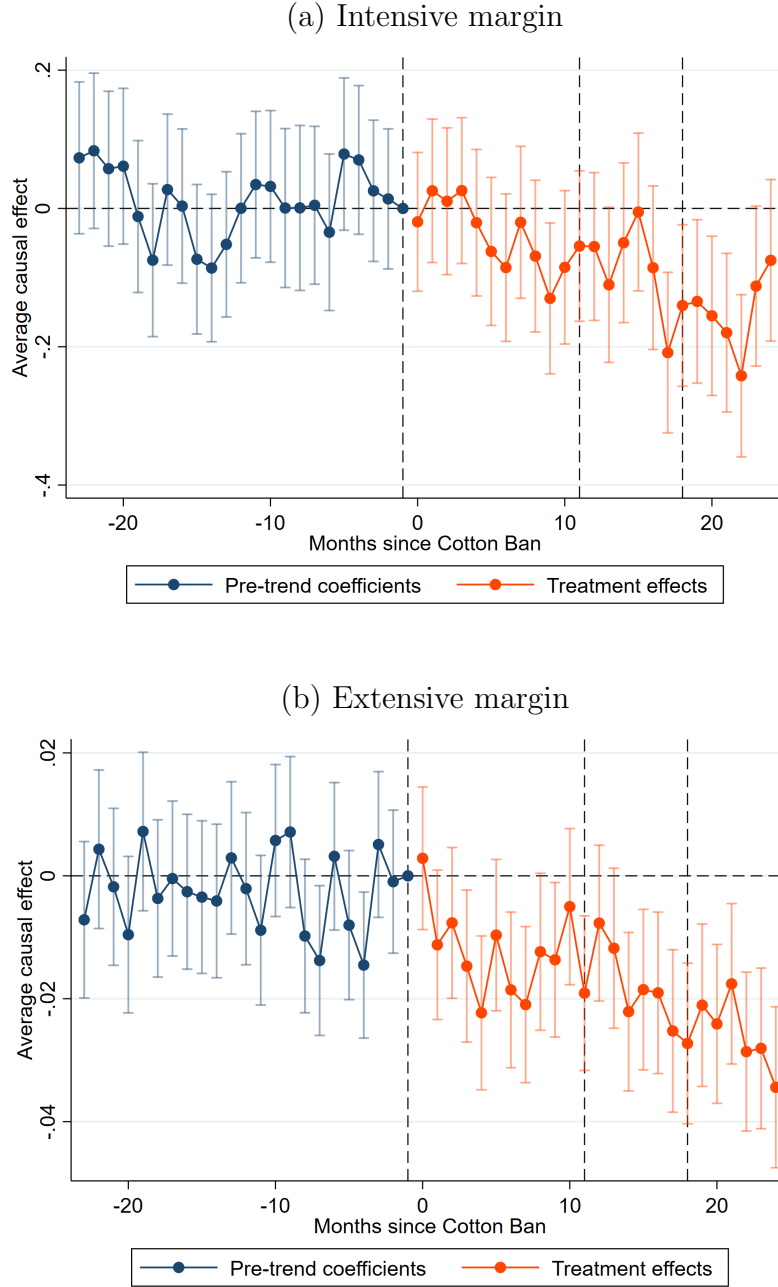
Notes: The figure shows the estimated coefficients for 47 individual interaction terms, along with the corresponding 95% confidence intervals, obtained by estimating equation (2). All regressions include product-destination, product-time, and destination-time fixed effects. Standard errors are clustered at the product-destination level. The vertical dashed lines indicate November 2020 (the baseline period, -1 , with values normalized at 0), November 2021, and June 2022, the three months before each of the three US Cotton Ban measures are enacted.

Figure 4: Effect of the US Cotton Ban on third country imports of Chinese cotton products



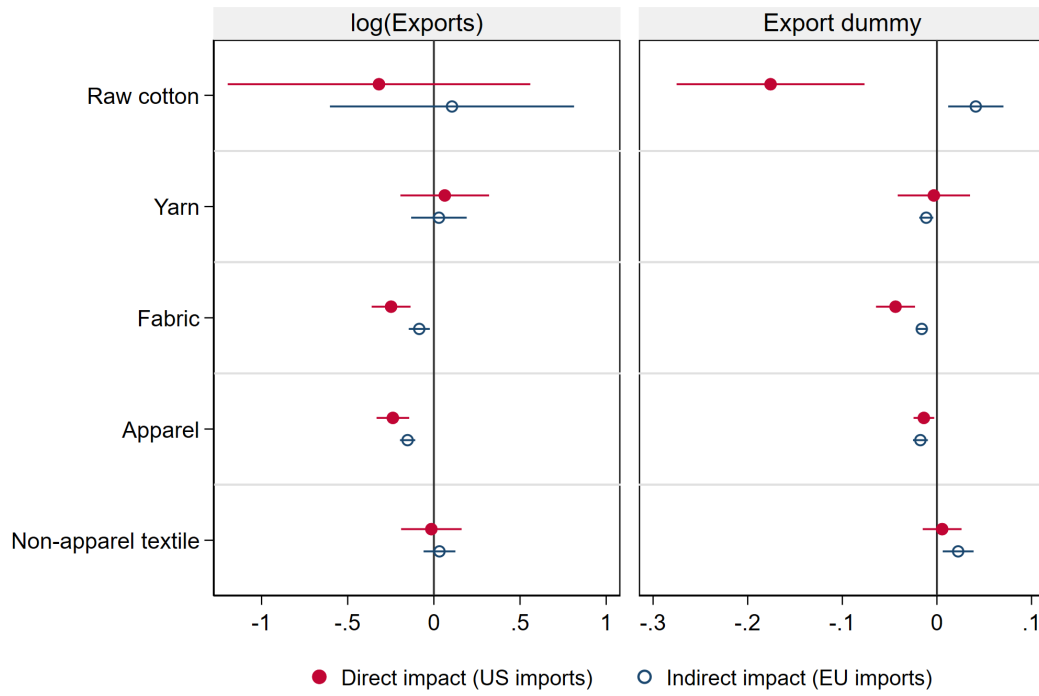
Notes: Point estimates and 95% confidence intervals of the β^D coefficient that captures the indirect effect of the US Cotton ban on imports in country D of cotton-containing products from China. Each coefficient is estimated using a separate regression for each destination, which always also incorporates an indicator for the direct effect on US imports and three sets of interaction fixed effects.

Figure 5: Event study of the indirect impact on EU imports



Notes: The figure shows the estimated coefficients for 47 individual interaction terms, along with the corresponding 95% confidence intervals, for the indirect effects on EU imports of cotton-related products that fall under the US Cotton Ban. All regressions also include 47 terms capturing the direct effect on US imports, as well as three sets of interaction fixed effects. Standard errors are clustered at the product-destination level. The vertical dashed lines indicate November 2020 (the baseline period, -1 , with values normalized at 0), November 2021, and June 2022, the three months before each of the three US Cotton Ban measures are enacted.

Figure 6: Heterogeneous impact of the Cotton Ban by product category



Notes: Coefficient estimates with 95% confidence intervals from separate regressions for each of the five product categories and both dependent variables. The sample period ranges from January 2019 to December 2022. Regressions include indicators for both the direct (US) and indirect (EU) effects and three sets of interaction fixed effects.

Table 1: Summary statistics on Chinese exports of cotton and control products (monthly, 2019)

	(1)		(2)		(3)	
	US cotton imports		EU cotton imports		Control imports	
	obs.	mean	obs.	mean	obs.	mean
Panel A: ALL						
Exports (USD 1,000)	2,856	3536	74,256	128	951,588	116
Export dummy	2,856	0.75	74,256	0.37	951,588	0.37
log(Exports)	2,141	12.03	27,669	9.62	356,499	9.45
Panel B: Raw cotton						
Exports (USD 1,000)	60	1	1,560	1	34,452	9
Export dummy	60	0.18	1,560	0.15	34,452	0.05
log(Exports)	11	7.61	238	6.14	1,855	8.11
Panel C: Yarn						
Exports (USD 1,000)	588	12	15,288	5	178,200	48
Export dummy	588	0.25	15,288	0.08	178,200	0.21
log(Exports)	145	9.32	1,243	8.02	37,179	9.37
Panel D: Fabric						
Exports (USD 1,000)	1,512	237	39,312	15	399,168	91
Export dummy	1,512	0.86	39,312	0.28	399,168	0.30
log(Exports)	1,293	10.64	11,186	8.71	118,423	9.74
Panel E: Apparel						
Exports (USD 1,000)	564	15010	14,664	574	250,668	218
Export dummy	564	0.99	14,664	0.84	250,668	0.59
log(Exports)	560	15.28	12,276	10.64	147,694	9.36
Panel F: Other textile						
Exports (USD 1,000)	132	9597	3,432	116	92,664	113
Export dummy	132	1.00	3,432	0.79	92,664	0.57
log(Exports)	132	15.14	2,726	9.84	52,637	9.14

Source: UN Comtrade

Notes: The number of HS 6-digit products included in each panel are: ALL (238), Raw (5), Yarn (49), Fabric (126), Apparel (47), Other textile (11). The EU encompasses all 26 member states, with the exception of Austria, which does not provide monthly statistics. We adapt the method of Carlson and Weaver (2022) to categorize the sample commodities into five different groups. Column (3) with “Control imports” includes: (i) all commodities imported by destinations other than the United States and the EU; (ii) non-cotton containing commodities imported by the United States and the EU.

Table 2: Summary statistics on Chinese provincial exports of cotton and control products (monthly, 2019)

	(1)		(2)		(3)	
	US cotton imports		EU cotton imports		Control imports	
	obs.	mean	obs.	mean	obs.	mean
Panel A: ALL						
Exports (USD 1,000)	82,476	136	2,144,376	4	32,676,156	6
Export dummy	82,476	0.16	2,144,376	0.03	32,676,156	0.03
log(Exports)	13,422	10.49	69,120	9.41	1,102,745	9.45
Panel B: Exports from Xinjiang						
Exports (USD 1,000)	2,844	2	73,944	0.2	1,126,764	5
Export dummy	2,844	0.01	73,944	0.00	1,126,764	0.01
log(Exports)	31	10.61	183	9.48	7,642	10.90
Panel C: Exports from Chinese provinces in 4th Quartile of Xinjiang sourcing						
Exports (USD 1,000)	19,908	318	517,608	10	7,887,348	14
Export dummy	19,908	0.31	517,608	0.07	7,887,348	0.07
log(Exports)	6,133	10.78	36,880	9.43	573,840	9.53

Source: China’s General Administration of Customs, Provincial trade data

Notes: The EU encompasses all 26 member states, excluding Austria for consistency with the UN Comtrade data. Column (3) with “Control imports” includes: (i) all commodities imported by destinations other than the United States and the EU; (ii) non-cotton containing commodities imported by the United States and the EU. We use the 2017 information from the interprovincial input–output table of Chen et al. (2023) to quantify the supply-chain linkages between Xinjiang and other Chinese provinces (excluding Tibet and Beijing). Chinese provinces (other than Xinjiang) are categorized into four groups based on the total value of agricultural and textile inputs in their local textile industry that is sourced from Xinjiang. Panel C shows summary statistics for the 4th Quartile, i.e. provinces above the 75th percentile.

Table 3: Direct impact of the Cotton Ban

	(1)	(2)	(3)	(4)
	log(Exports)	log(Exports)	Export dummy	Export dummy
Ban^{US}	-0.211*** (0.044)		-0.030*** (0.008)	
$Ban^{US} \times first$		-0.169*** (0.044)		-0.018** (0.008)
$Ban^{US} \times second$		-0.182*** (0.059)		-0.020* (0.011)
$Ban^{US} \times UFLPA$		-0.331*** (0.064)		-0.067*** (0.013)
Observations	1,914,465	1,914,465	4,844,448	4,844,448
Product-destination FE	✓	✓	✓	✓
Product-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓	✓	✓

Notes: Estimation results for equation (1); the units of observation are imports from China at the product-destination-month; the sample period ranges from January 2019 to December 2022. Ban^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported from China to the United States after the implementation of the Cotton Ban in December 2020. The three interaction terms in columns (2) and (4) represent different time periods: (1) *first* is the period between December 2020 and November 2021, the first year of the Cotton Ban; (2) *second* is the period between December 2021 and June 2022, the second year of the Cotton Ban, when the *UFLPA* was enacted but not yet in force; and (3) *UFLPA* is the period from July 2022 onward when the *UFLPA* is in effect. Standard errors in parentheses are clustered at the product-destination level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table 4: Direct impact of the Cotton Ban on provincial exports

	(1)	(2)	(3)	(4)
	Destination-month FE		Product-destination-month FE	
	log(Exports)	Export Dummy	log(Exports)	Export Dummy
$Ban^{US} \times \text{Xinjiang}$	-1.105*** (0.328)	-0.024*** (0.003)	-1.732*** (0.455)	-0.013*** (0.004)
$Ban^{US} \times \text{Q4 Xinjiang sourcing}$	-0.271*** (0.041)	-0.012*** (0.003)	-0.750*** (0.278)	-0.001 (0.003)
$Ban^{US} \times \text{Q3 Xinjiang sourcing}$	-0.113** (0.049)	-0.024*** (0.003)	-0.684** (0.279)	-0.013*** (0.003)
$Ban^{US} \times \text{Q2 Xinjiang sourcing}$	-0.037 (0.062)	-0.012*** (0.002)	-0.591** (0.282)	-0.001 (0.003)
$Ban^{US} \times \text{Q1 Xinjiang sourcing}$	0.640** (0.287)	-0.011*** (0.002)		
Observations	4,750,327	139,612,032	4,173,657	139,612,032
Product-province-destination FE	✓	✓	✓	✓
Product-province-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓		
Product-destination-month FE			✓	✓

Notes: Estimation results for equation (3) with Chinese provinces as origins (exporters); the units of observation are exports of China at the product-province-destination-month; the sample period ranges from January 2019 to December 2022. Ban^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported from any Chinese province to the United States after the implementation of the Cotton Ban in December 2020. The first interaction term is a dummy for direct exports from Xinjiang province. The next four interactions are dummies for the quartile each province is assigned to in terms of sourcing agricultural and textile inputs from Xinjiang. For the results in columns (3) and (4), the average effect is absorbed by the full set of interaction fixed effects, and $Q1$ is omitted as a reference group. Standard errors in parentheses are clustered at the product-province-destination level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table 5: Direct impact of the Cotton Ban according to countries' input sourcing from China

	(1)	(2)	(3)	(4)
	Destination-month FE		Product-destination-month FE	
	log(Exports)	Export dummy	log(Exports)	Export dummy
$Ban_{pdt}^{US} \times \text{China} (\equiv Ban^{US})$	-0.368*** (0.039)	-0.050*** (0.007)	-0.460*** (0.051)	-0.042*** (0.008)
$Ban_{pdt}^{US} \times \text{Q4 China sourcing}$	0.024 (0.029)	-0.005 (0.004)	-0.067 (0.046)	0.002 (0.004)
$Ban_{pdt}^{US} \times \text{Q3 China sourcing}$	0.053 (0.036)	-0.011*** (0.004)	-0.026 (0.052)	-0.004 (0.004)
$Ban_{pdt}^{US} \times \text{Q2 China sourcing}$	0.056* (0.031)	-0.004 (0.003)	-0.031 (0.048)	0.003 (0.004)
$Ban_{pdt}^{US} \times \text{Q1 China sourcing}$	0.091*** (0.034)	-0.007** (0.003)		
Observations	9,667,684	100,964,448	8,839,862	100,964,448
Product-origin-destination FE	✓	✓	✓	✓
Product-origin-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓		
Product-destination-month FE			✓	✓

Notes: Estimation results for equation (3) with third countries as origins (exporters); the units of observation are product-origin-destination-months; the sample period ranges from January 2019 to December 2022. Twenty-one origin countries are China, Belgium, Bulgaria, Cambodia, Canada, Colombia, Dominican Republic, France, Germany, Guatemala, India, Indonesia, Italy, Japan, Malaysia, Mexico, Nicaragua, Pakistan, Portugal, Thailand, and Vietnam. Ban_{pm}^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported (from any origin country) to the United States after the implementation of the Cotton Ban in December 2020. The first interaction term is a dummy for direct exports from China. The next four interactions are dummies for the quartile each origin country (apart from China) is assigned to based on its total imports of all cotton products from China in 2017 and 2018. In columns (3) and (4), the average effect is absorbed by the full set of interaction fixed effects, and Q1 is omitted as a reference group. Standard errors in parentheses are clustered at the product-origin level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table 6: Indirect impact of the Cotton Ban: Chinese exports to the EU

	(1)	(2)	(3)	(4)
	log(Exports)	log(Exports)	Export dummy	Export dummy
Ban^{US}	-0.234*** (0.044)		-0.034*** (0.008)	
$Ban^{US} \times first$		-0.182*** (0.045)		-0.020** (0.008)
$Ban^{US} \times second$		-0.208*** (0.060)		-0.024** (0.011)
$Ban^{US} \times UFLPA$		-0.372*** (0.065)		-0.072*** (0.013)
Ban^{EU}	-0.092*** (0.021)		-0.015*** (0.003)	
$Ban^{EU} \times first$		-0.051** (0.021)		-0.010*** (0.003)
$Ban^{EU} \times second$		-0.103*** (0.027)		-0.016*** (0.003)
$Ban^{EU} \times UFLPA$		-0.160*** (0.030)		-0.023*** (0.004)
Observations	1,914,465	1,914,465	4,844,448	4,844,448
Product-destination FE	✓	✓	✓	✓
Product-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓	✓	✓

Notes: Estimation results for equation (1); the units of observation are imports from China at the product-destination-month; the sample period ranges from January 2019 to December 2022. Ban^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported from China to the United States after the implementation of the Cotton Ban in December 2020. Ban^{EU} is a comparable indicator, but for Chinese imports into any of the 26 EU countries; the sample omits Austria which does not provide monthly trade information and the UK is not included in the EU. The three interaction terms in columns (2) and (4) represent different time periods: (1) *first* is the period between December 2020 and November 2021, the first year of the Cotton Ban; (2) *second* is the period between December 2021 and June 2022, the second year of the Cotton Ban, when the *UFLPA* was enacted but not yet in force; and (3) *UFLPA* is the period from July 2022 onward when the *UFLPA* is in effect. Standard errors in parentheses are clustered at the product-destination level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table 7: Indirect impact of the Cotton Ban: Price versus quantity

	(1) log(Exports)	(2) log(Price)	(3) log(Quantity)
Ban^{US}	-0.234*** (0.044)	0.030 (0.024)	-0.253*** (0.055)
Ban^{EU}	-0.092*** (0.021)	0.000 (0.010)	-0.092*** (0.024)
Observations	1,914,465	1,866,363	1,866,364
Product-destination FE	✓	✓	✓
Product-month FE	✓	✓	✓
Destination-month FE	✓	✓	✓

Notes: Estimation results for equation (1) using three different dependent variables. The units of observation are imports from China at the product-destination-month; the sample period ranges from January 2019 to December 2022. Ban^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported from China to the United States after the implementation of the Cotton Ban in December 2020. Ban^{EU} is a comparable indicator, but for Chinese imports into any of the 26 EU countries. The sample omits Austria which does not provide monthly trade information and the UK is not included in the EU. Standard errors in parentheses are clustered at the product-destination level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table 8: Indirect impact of the Cotton Ban: Re-export versus reputation considerations

	(1)	(2)	(3)	(4)
	log(Exports)	log(Exports)	Export dummy	Export dummy
Ban^{US}	-0.234*** (0.044)	-0.234*** (0.044)	-0.034*** (0.008)	-0.034*** (0.008)
Ban^{EU}	-0.118*** (0.023)	-0.031 (0.025)	-0.015*** (0.003)	-0.019*** (0.003)
$Ban^{EU} \times \text{High US exports}$	0.127*** (0.042)		-0.001 (0.005)	
$Ban^{EU} \times \text{High HR aware}$		-0.153*** (0.036)		0.012** (0.005)
Observations	1,914,465	1,914,465	4,844,448	4,844,448
Product-destination FE	✓	✓	✓	✓
Product-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓	✓	✓

Notes: Estimation results for equation (1) with additional interaction terms; the units of observation are imports from China at the product-destination-month; the sample period ranges from January 2019 to December 2022. Ban^{US} is the coefficient on the triple interaction indicator taking a value of one if a commodity contains cotton and was exported from China to the United States after the implementation of the Cotton Ban in December 2020. Ban^{EU} is a comparable indicator, but for Chinese imports into any of the 26 EU countries. The sample omits Austria which does not provide monthly trade information and the UK is not included in the EU. The interaction term “High US exports” is an indicator that equals one for country-product combinations where the EU country has a US export share for that specific cotton-containing product that exceeds the 75th percentile (across EU countries). The “High HR search” indicator equals one for EU countries with a relative Google keywords search intensity for human rights crises that exceeds the median (across EU countries). Standard errors in parentheses are clustered at the product-destination level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Appendix A Sample of commodities

Table A.1: Number of products in the sample (treated products in parenthesis)

HS 2-digit	Description	Raw	Yarn	Fabrics	Apparel	Other textiles
50	Silk	3	2	4	0	0
51	Wool, fine or coarse animal hair; horsehair yarn and woven fabric	12	15	11	0	0
52	Cotton	5 (5)	49 (49)	70 (70)	0	0
53	Vegetable textile fibres; paper yarn and woven fabrics of paper yarn	9	7	7	0	0
54	Man-made filaments	0	36	34 (3)	0	0
55	Man-made staple fibres	0	24	84 (31)	0	0
56	Wadding, felt and nonwovens, special yarns; twine, cordage, ropes and cables and articles thereof	0	14	17 (1)	0	0
57	Carpets and other textile floor coverings	0	0	0	0	23
58	Fabrics; special woven fabrics, tufted textile fabrics, lace, tapestries, trimmings, embroidery	0	0	37 (9)	0	0
59	Textile fabrics; impregnated, coated, covered or laminated; textile articles of a kind suitable for industrial use	0	0	24	0	0
60	Fabrics; knitted or crocheted	0	0	44 (12)	0	0
61	Apparel and clothing accessories; knitted or crocheted	0	0	0	106 (25)	0
62	Apparel and clothing accessories; not knitted or crocheted	0	0	0	104 (22)	0
63	Textiles, made up articles; sets; worn clothing and worn textile articles; rags	0	3	4	1	47 (11)
65	Headgear and parts thereof	0	0	0	0	8
TOTAL		29(5)	150 (49)	336 (126)	211 (47)	78 (11)

Source: World Integrated Trade Solution (WITS) of the World Bank (2024), accessible at <https://wits.worldbank.org/trade/country-byhs6product.aspx?lang=en>.

Notes: Products are defined at the HS 6-digit level. The number of cotton-containing commodities are in parentheses. We use the method of Carlson and Weaver (2022) to categorize 801 commodities at the HS 6-digit level into five groups. Three control commodities in 63 are classified as both *Yarn* and *Fabrics*.

Appendix B Province ranking based on intensity of input sourcing from Xinjiang

We use the supply chain links between Chinese provinces and Xinjiang observed in the interprovincial Input-Output table of Chen et al. (2023) to classify provinces into four groups. For each province, we calculate for the two sectors of ‘Manufacture of textiles’ and ‘Manufacture of textile wearing apparel, footwear, leather, fur, feather and its products’ how much of their inputs come from the Xinjiang sector of ‘Farming, forestry, animal production and fishery’ or from either of the two Xinjiang sectors mentioned above. Results in Table B.1 illustrate that the bulk of output from the first two sectors is used as input in the textile sectors of Xinjiang or other Chinese provinces, but only a small fraction of the third sector.

Naturally, these broadly defined categories will encompass some non-cotton inputs from Xinjiang, but we do not observe supply linkages at a more detailed level of industry or product aggregation. From the Xinjiang Statistical Yearbook we know that cotton accounts for 24% of total output in the Xinjiang agricultural sector and we have multiplied the inputs from that sector by 0.24 before showing them in Table B.1 and adding them to the inputs of the other two sectors.

We rank all Chinese provinces according to their total sourcing from Xinjiang, shown in Figure B.1, and categorize them into four quartiles based on that index.

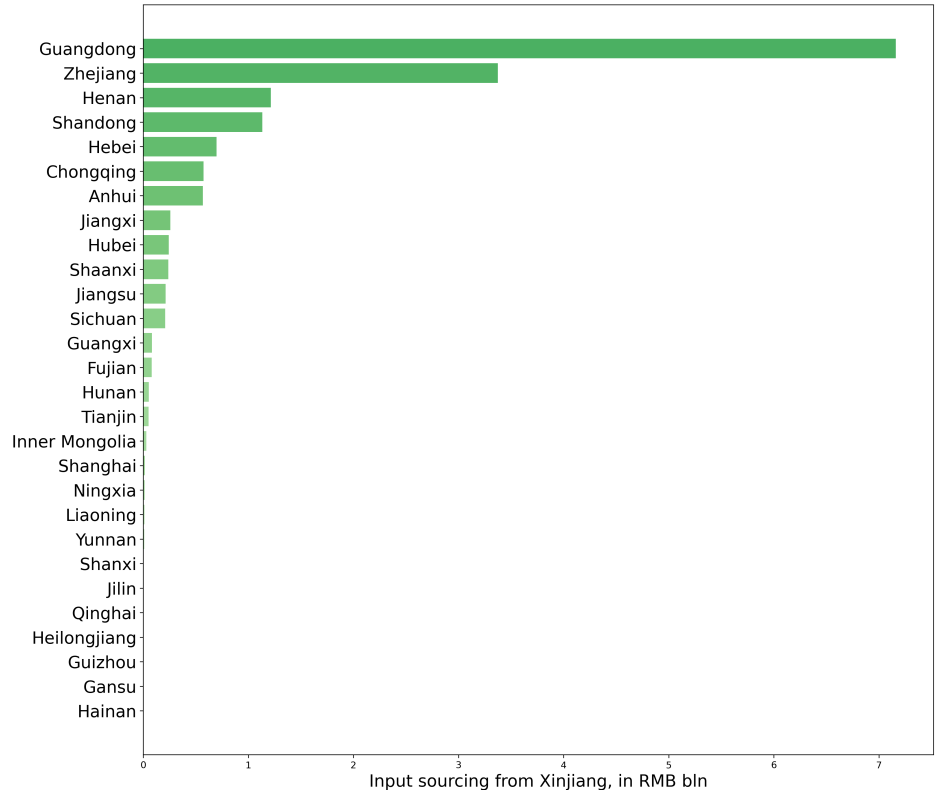
Table B.1: Destination of the output of cotton-related sectors in Xinjiang

	(1)	(2)	(3)	(4)	(5)	(6)
	Intermediate use		Domestic consumption		Exports	Total
(in millions, RMB)	Xinjiang	Others	Xinjiang	Others		
Xinjiang Agricultural	23,167	16,927	11,520	17,794	663	70,071
	(33%)	(24%)	(16%)	(25%)	(1%)	(100%)
Xinjiang Textile	6,547	19,747	1,105	1,078	10,506	38,983
	(17%)	(51%)	(3%)	(3%)	(27%)	(100%)
Xinjiang Garment	249	0.3	1,152	0.2	3,820	5,222
	(5%)	(0%)	(22%)	(0%)	(73%)	(100%)

Source: Calculations based on China interprovincial input-output (IPIO) database for 2017 of Chen et al. (2023).

Notes: Columns (1)-(5) show the value and share of output of sector i in Xinjiang that is used as intermediate input or domestic consumption (consumption + investment) in Xinjiang or any other Chinese province, or is exported.

Figure B.1: Chinese provinces ranked by cotton-related input sourcing from Xinjiang



Source: Calculations based on China interprovincial input–output database 2017 of Chen et al. (2023).

Notes: Values indicate the cotton-related intermediate inputs (3 input-output sectors) from Xinjiang that are used in textile-related sectors (2 input-output sectors) of each province. Tibet and Beijing are excluded from the analysis.

Appendix C Country ranking based on attention to human rights crises

Google Trends makes information regarding the online search activity for a broad range of keywords available to researchers by country and time period. It does not release the absolute number of searches, but employs a two-step normalization process. First, the daily search volume for each keyword is adjusted by the total daily search volume across all keywords in a country. Second, the results are further normalized by the maximum daily search volume within a sample period. If a search term consists of two keywords, the normalization is relative to the maximum daily search intensity of either of the two keywords. The resulting relative search intensity for each keyword or group of keywords falls within a range of 0 to 100.

We collect data on the relative search intensity corresponding to several historical human rights crises that occurred in the 2010s. Each of these crises occurred in different regions and have drawn condemnation from the European Parliament. Rather than working with individual keywords or search terms, we identify so-called “topics” associated with an incident. Topics are conceptual grouping of related terms, that even include words in different languages. They are classified by Google Trends and are the building blocks we work with. This approach facilitates capturing the intensity of search over all relevant search terms associated with an event. In some cases we pick “ethnic groups”, which is a group of related keywords much like a topic and is also defined by Google Trends. In a time period surrounding human rights violations that involve a certain ethnic group, we assume the search activity in EU countries pertain to that specific event. We select “Traffic (Topic)” as a control search topic to normalize the average search intensity over topics separately for each EU country.

The first three lines in Table C.1 lists three human rights crises that we could clearly associate with a Google Trends-defined topic and that occurred prior to our study period.

Table C.1: Keywords for selected human right crises

Human rights crisis	Google Trends topic	Peak of global search	2-month search period
Boko Haram kidnappings	Boko Haram (Topic)	May 12, 2014	April 13 – June 11
Yazidi genocide	Yazidis (People)	Aug 10, 2014	July 12 – Sept. 9
Rohingya genocide	Rohingya people (Ethnic group)	Sep 6, 2017	Aug. 8 – Oct. 6
2019 Turkish offensive into north-eastern Syria	Kurds (Ethnic group)	Oct 10, 2019	Sept. 11 – Nov. 9
Persecution of Uyghurs in China	Uyghurs (Ethnic group)	Dec 22, 2019	Nov. 23 – Jan. 21

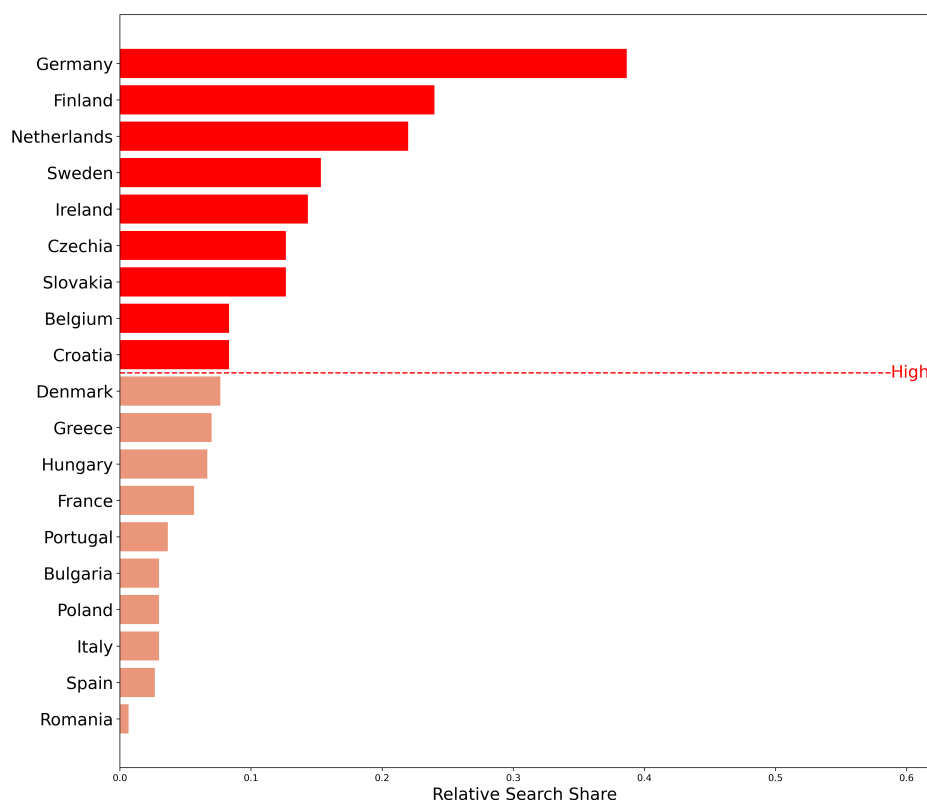
Source: “Topic” definitions and date of peak global search are obtained from the Google Trends database.

They include the attacks on schools and the kidnappings of schoolgirls by the jihadist militant group Boko Haram in Nigeria, the Yazidi genocide and refugee crisis caused by Islamic State in Northern Iraq, and the genocide against the Rohingya people by the army in Myanmar.

We calculate a search intensity index for each EU country by averaging the 2-month relative search share (against “Traffic”). Search data for all three crises in Cyprus, Estonia, Latvia, Lithuania, Luxembourg, Malta, and Slovenia are labeled as low volumes searches by Google Trends and we do not calculate the index for them. Figure C.1 shows the ranking across EU countries and the dividing line for the median index.

Robustness checks using search volumes over a single month or using two additional crises, including the Chinese treatment of Uyghur minorities in Xinjiang, yield a very similar ranking.

Figure C.1: European countries ranked by relative online search intensity for human rights crises



Notes: The search intensity regarding three human rights crises is calculated using data from Google Trends. These crises are (1) the Boko Haram kidnappings in 2014, (2) the Yazidi genocide in 2014; and (3) the Rohingya genocide in 2017. The two-month relative search intensities on each of these three Google Trends-generated “topics” is normalized by the “Traffic” topic in each EU country. Seven EU countries are excluded because Google designates them as having too low search volume. Furthermore, Austria is not included in our sample.

Appendix D Country rankings using trade data

There are two ways for countries to be highly exposed to the US Cotton Ban. First, countries can source lots of cotton products from China, or second, countries can export lots of cotton products to the United States.

We first investigate whether the reduction of US imports is heterogeneous according to exporters' sourcing from China. In the analysis reported in Table 5, we extend the baseline three-dimensional balanced panel of Chinese exports with comparable information for 20 additional origin countries. Panel A of Table D.1 reports these countries' rankings as importers of cotton products from China and as exporters of cotton products to the United States in 2017 and 2018. The countries are chosen to all have high US export exposure, but differing in their reliance on Chinese imports. They range from the 1st to the 55th largest exporter (to the United States) and from the 2nd to the 76th largest importer (from China).

In a robustness check reported in Table E.3, we add six additional origin countries to the four-dimensional sample. They are reported in Panel B of Table D.1. The full sample of 26 countries used in that analysis covers all top-30 cotton exporting countries (to the United States) that have complete monthly trade data over the sample period.

In a second robustness check, we estimate a triple-difference specification similar to the baseline analysis. In this case, we construct a three-dimensional balanced panel of US imports from all origins, rather than Chinese exports to all destinations. We calculate the total

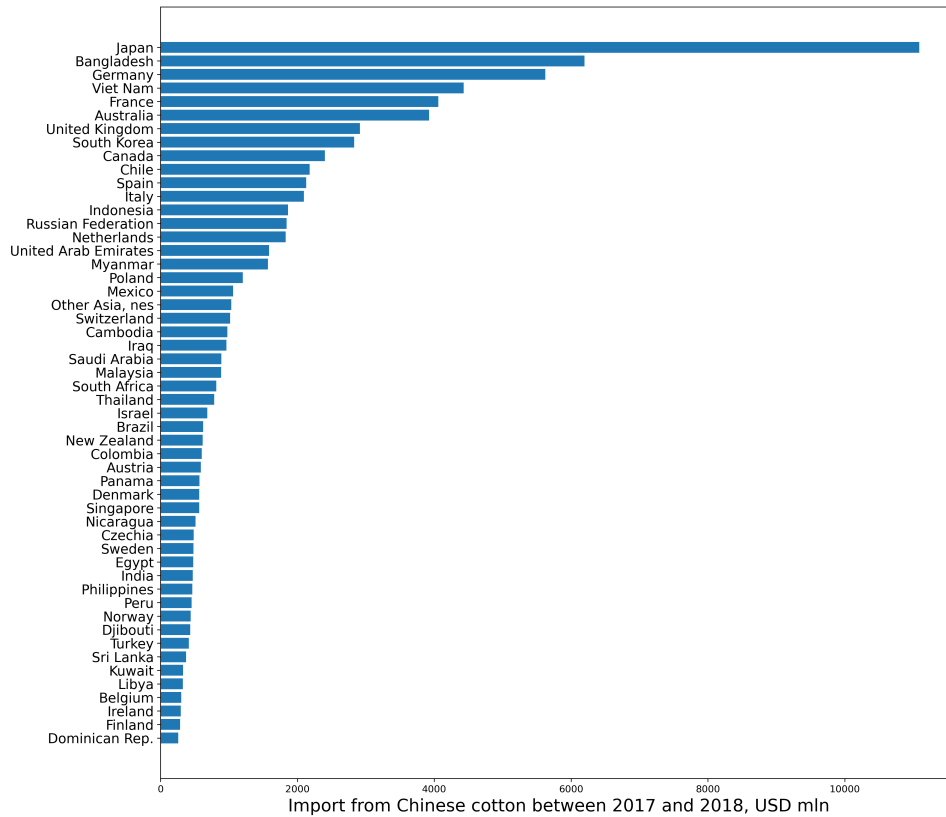
Table D.1: Rankings of origin countries

Origin	Ranking as importer from China	Ranking as exporter to US	Origin	Ranking as importer from China	Ranking as exporter to US
Panel A: Twenty selected origin countries (used in Table 5)					
Belgium	41	55	Indonesia	13	5
Bulgaria	76	47	Italy	12	16
Cambodia	21	8	Japan	2	28
Canada	9	26	Malaysia	22	21
Colombia	28	29	Mexico	19	6
Dominican Rep.	44	17	Nicaragua	32	9
France	5	33	Pakistan	57	4
Germany	3	44	Portugal	63	23
Guatemala	60	12	Thailand	24	19
India	36	2	Vietnam	4	1
Panel B: Six additional origin countries (added in Table E.3)					
Egypt	35	18	Philippines	37	22
El Salvador	49	10	South Korea	8	24
Mauritius	55	27	Turkey	40	13

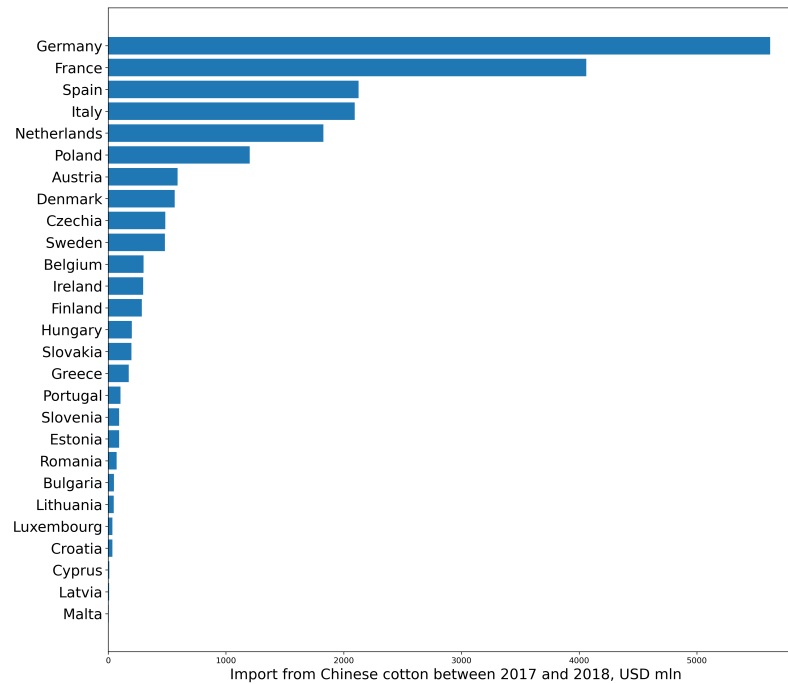
Notes: Panel A displays the 20 origin countries (in addition to China) used for the analysis reported in Table 5 and Panel B the six additional countries added to the sample for the analysis in Table E.3. The rankings are calculated based on imports and exports of cotton products in 2017 and 2018.

Figure D.1: Country rankings by cotton input sourcing from China

(a) All countries (4th quartile)



(b) EU countries



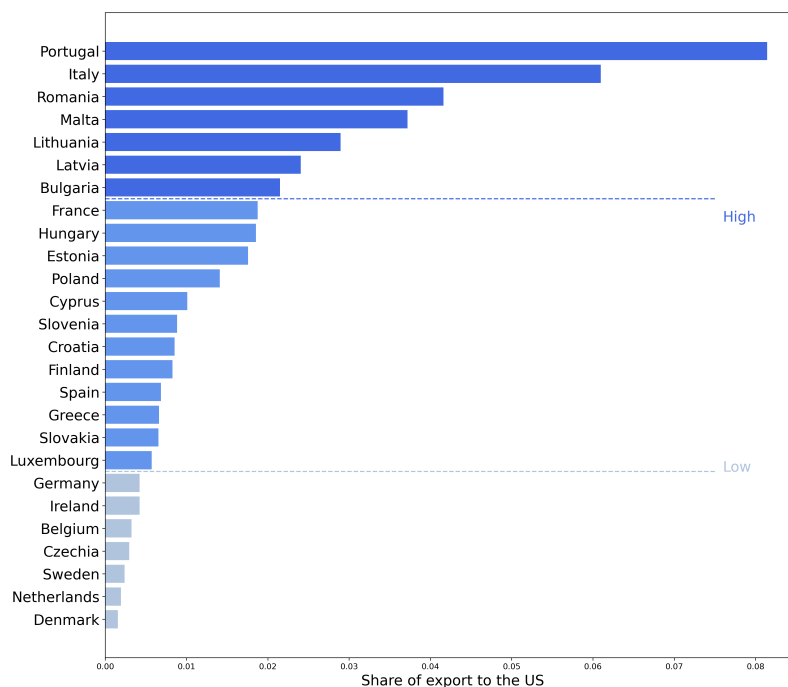
Source: Calculations based on UN Comtrade data.

Notes: Import values of cotton and cotton products from China in 2017 and 2018. Austria is excluded.

import value of cotton-related products from China for each of 207 origin countries in 2017 and 2018. For 33 origin countries that do not report annual import data in UN Comtrade, we rely on the Chinese export data. Countries are categorized into four quartiles according to their extent of cotton sourcing from China and origin countries in the top quartile are shown in panel (a) of Figure D.1. A similar ranking based only on EU countries is shown in panel (b).

Second, we also investigate heterogeneity of the indirect effects of the Cotton ban. Countries with higher own exports of cotton-related products to the United States may protect their access to the large US market by changing to alternative sources of cotton-related inputs than China. Specifically for EU countries, we measure what fraction of each country's exports of cotton textiles and garments (excluding raw cotton) in 2017 and 2018 are directed to the United States. A higher ratio reflects stronger dependence on the US market and, consequently, greater exposure to the indirect effects of the Cotton Ban. Austria is excluded from the sample, while for Luxembourg the measure is based only on 2018, as only 10 observations are reported in 2017. The ranking is shown in Figure D.2 and it is used to classify EU countries in three terciles, as indicated.

Figure D.2: EU countries ranked by cotton exports to the United States



Source: Calculated based on UN Comtrade

Notes: Share of US exports in total exports of cotton-containing textiles and garments for 2017 and 2018. Austria is excluded.

Appendix E Additional results

Table E.1: Direct impact of the Cotton Ban: PPML estimator

	(1)	(2)
	Exports	Exports
Ban^{US}	-0.105*** (0.031)	
$Ban^{US} \times first$		-0.052 (0.037)
$Ban^{US} \times second$		-0.092* (0.041)
$Ban^{US} \times UFLPA$		-0.224*** (0.046)
Observations	3,647,662	3,647,662
Destination-Product FE	✓	✓
Product-Month FE	✓	✓
Destination-Month FE	✓	✓

Notes: Results estimated with Poisson Pseudo Maximum Likelihood, using the level of exports as dependent variable. Ban^{US} is an indicator that equals one if a commodity contains cotton, the export destination is the United States, and the Cotton Ban is in effect (from December 2020 onward). *first* is an interaction dummy for the period between December 2020 and November 2021, *second* for the period between December 2021 and June 2022, and *UFLPA* for the period from July 2022 onward. Standard errors, clustered at product-destination level, are reported in parentheses with ***, **, and * represent significance at the 1%, 5%, and 10% level, respectively.

Table E.2: Direct impact of the Cotton Ban: Replication with China's Customs data

	(1)	(2)	(3)	(4)
	Province-level data (<i>podt</i>)		National (aggregated) data (<i>pdt</i>)	
	log(Exports)	Export dummy	log(Exports)	Export dummy
Ban^{US}	-0.169*** (0.030)	-0.015*** (0.002)	-0.296*** (0.062)	-0.002 (0.010)
Observations	4,750,327	139,612,032	1,469,040	4,814,208
Product-province-destination FE	✓	✓		
Product-destination FE			✓	✓
Destination-Month FE	✓	✓	✓	✓
Product-province-month FE	✓	✓		
Product-month FE			✓	✓

Notes: Results are based on the provincial-level data from China's Customs. In columns (1) and (2), the units of observations are product-province-destination-months and standard error are clustered at the product-province-destination level. In columns (3) and (4), the data is aggregated to the national level across provinces and it uses the same triple-difference estimator of the benchmark analysis. Standard errors are clustered at the product-destination level. Clustered standard errors are reported in parentheses with ***, **, and * represent significance at the 1%, 5%, and 10% level, respectively.

Table E.3: Direct impact of the Cotton Ban according to countries' input sourcing from China: Robustness check using 27 origin countries

	(1)	(2)	(3)	(4)
	Destination-month FE		Product-destination-month FE	
	log(Exports)	Export dummy	log(Exports)	Export dummy
$Ban_{pdt}^{US} \times \text{China} (\equiv Ban^{US})$	-0.379*** (0.039)	-0.049*** (0.007)	-0.475*** (0.049)	-0.042*** (0.008)
$Ban_{pdt}^{US} \times \text{Q4 China sourcing}$	0.003 (0.024)	-0.006** (0.003)	-0.099** (0.041)	0.000 (0.004)
$Ban_{pdt}^{US} \times \text{Q3 China sourcing}$	0.014 (0.044)	-0.009*** (0.003)	-0.047 (0.057)	-0.002 (0.004)
$Ban_{pdt}^{US} \times \text{Q2 China sourcing}$	0.125*** (0.028)	0.002 (0.003)	0.019 (0.044)	0.004 (0.004)
$Ban_{pdt}^{US} \times \text{Q1 China sourcing}$	0.081** (0.031)	-0.007*** (0.002)		
Observations	11,326,141	129,800,448	10,577,972	129,800,448
Product-origin-destination FE	✓	✓	✓	✓
Product-origin-month FE	✓	✓	✓	✓
Destination-month FE	✓	✓		
Product-destination-month FE			✓	✓

Notes: Estimation results for equation (3) with third countries as origins (exporters); the units of observation are product-origin-destination-months; the sample period ranges from January 2019 to December 2022. Twenty-seven origin countries are China, Belgium, Bulgaria, Cambodia, Canada, Colombia, Dominican Republic, Egypt, El Salvador, France, Germany, Guatemala, India, Indonesia, Italy, Japan, Malaysia, Mauritius, Mexico, Nicaragua, Pakistan, Philippines, Portugal, South Korea, Turkey, Thailand, and Vietnam. This includes six more countries than in Table 5, now covering all top-30 exporters to the United States that report a complete monthly trade data for the full sample period. The first interaction term is a dummy for direct exports from China. The next three interactions are dummies for the quartile each origin country (apart from China) is assigned to based on its total imports of raw cotton, cotton yarns, and cotton fabrics from China over the 2017–2018 period. The results in columns (3) and (4) exclude raw cotton imports from the sample. Note that the average effect is absorbed by the full set of interaction fixed effects, and Q1 is omitted as a reference group. Standard errors in parentheses are clustered at the product-origin level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.

Table E.4: Direct impact of the Cotton Ban according to countries' input sourcing from China: Triple-difference analysis using only US imports

	(1)	(2)	(3)	(4)
	All		Excluding Raw Materials	
	log(Exports)	Export dummy	log(Exports)	Export dummy
$Ban_{pt}^{US} \times \text{China} (\equiv Ban^{US})$	-0.243** (0.121)	-0.037*** (0.008)	-0.245** (0.121)	-0.033*** (0.008)
$Ban_{pt}^{US} \times \text{Q4 China sourcing}$	0.049 (0.116)	-0.007*** (0.001)	0.049 (0.116)	-0.007*** (0.001)
$Ban_{pt}^{US} \times \text{Q3 China sourcing}$	0.036 (0.117)	-0.002* (0.001)	0.036 (0.117)	-0.002** (0.001)
$Ban_{pt}^{US} \times \text{Q2 China sourcing}$	0.036 (0.124)	-0.000 (0.001)	0.036 (0.124)	-0.001 (0.001)
Observations	611,433	7,124,544	608,390	6,865,632
Product-origin FE	✓	✓	✓	✓
Product-month FE	✓	✓	✓	✓
Origin-month FE	✓	✓	✓	✓

Notes: Estimation results for equation (3) with third countries as origins (exporters); the units of observation are US imports by product-origin-months; the sample period ranges from January 2019 to December 2022. Ban_{pm}^{US} is a dummy taking a value of one if a commodity contains cotton and was exported (from any origin country) to the United States after the implementation of the region-wide cotton ban in December 2020. The first interaction term is a dummy for direct exports from China. The next three interactions are dummies for the quartile each origin country (185 countries apart from China) is assigned to based on its total imports cotton products from China over the 2017–2018 period. The results in columns (3) and (4) exclude raw cotton imports from the sample. Note that the average effect is absorbed by the full set of interaction fixed effects, and Q1 is omitted as a reference group. Standard errors in parentheses are clustered at the product-origin level, with ***, **, and * representing significance at the 1%, 5%, and 10% level.